

Sharp Asymptotics of Learning Multi-Index Models via Principal Hessian Directions

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Abstract

This paper presents an exact asymptotic analysis of a multi-rank spectral method for estimating an r -dimensional subspace from nonlinear measurements. We study the method, inspired by Principal Hessian Directions, in a high-dimensional regime where the number of measurements n and the signal dimension d grow to infinity with a fixed ratio $\alpha = n/d$. Our primary contribution is a precise characterization of the method's performance under a generalized linear sensing model. The analysis reveals a sharp phase transition phenomenon: we derive a set of deterministic fixed-point equations whose solutions show that a non-trivial alignment between the estimated and true subspaces emerges only above a critical threshold. Our formulae precisely quantify the inner products between the target signals and the eigenvectors, demonstrating that signal information is captured by both the leading and trailing eigenvectors of the data matrix. These results extend the prior theory for single-signal recovery (rank-1) to the general multi-signal (rank- r) setting. The accuracy of our theoretical predictions is confirmed by numerical simulations for problems such as multiplexed phase retrieval and learning shallow neural networks.

Keywords: Spectral methods, phase retrieval, subspace estimation, nonconvex optimization, phase transition, effective dimension reduction

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1 Introduction

1.1 Model

We consider the problem of estimating a low-rank subspace from nonlinear measurements. Specifically, let $\xi_1, \dots, \xi_r \in \mathbb{R}^d$ be a basis spanning a rank- r subspace that we want to estimate, and let $\{\mathbf{a}_i \in \mathbb{R}^d\}_{1 \leq i \leq n}$ be a collection of sensing vectors. Given $\{\mathbf{a}_i^\top \xi_1, \dots, \mathbf{a}_i^\top \xi_r\}$, the i th measurement $y_i \in \mathbb{R}$ is drawn independently from

$$y_i \sim f(y \mid \mathbf{a}_i^\top \xi_1, \dots, \mathbf{a}_i^\top \xi_r), \quad 1 \leq i \leq n, \quad (1)$$

where $f(\cdot \mid \cdot)$ is a conditional density function modeling the sensing process. We seek to estimate the underlying subspace spanned by $\{\xi_j\}_{1 \leq j \leq r}$ from the set of measurements $\{y_i\}_{1 \leq i \leq n}$. Since we consider the general sensing model in (1), we can always use QR-decomposition and adapt $f(\cdot \mid \cdot)$ accordingly to make sure that our target signals $\{\xi_1, \dots, \xi_r\}$ are orthonormal. This task arises in many important problems in statistical signal processing and machine learning. Examples include effective dimension reduction in exploratory data analysis [1, 2], covariance sketching [3, 4], coherence retrieval [5, 6], multiplexed imaging [7], and the learning of shallow neural networks [8, 9].

The intrinsic non-convexity of the general problem (1) presents challenges in developing efficient algorithms for the accurate recovery of target signals. For instance, in the specific case when $r = 1$, a well-studied problem known as phase retrieval [10] has the following formulation:

$$y_i = (\mathbf{a}_i^\top \xi)^2 + \epsilon_i$$

where $\{\epsilon_i\}$ are some noise. The work in [11–14] employs semidefinite programming which has guaranteed convergence but requires substantial computation and memory. This limitation motivates the development of various iterative algorithms later, such as alternating minimization [15, 16], Wirtinger-Flow algorithms [17–21], iterative projections [22, 23], and linear programming [24–27]. When considering the case $r > 1$, a widely studied problem is the mixture of linear regression, where

$$y_i = \sum_{j=1}^r \beta_j (\mathbf{a}_i^\top \xi_j) + \epsilon_i$$

for some noise $\{\epsilon_i\}$. A series of iterative methods are proposed for this problem, such as the expectation-maximization (EM) algorithms [28, 29] and alternating minimization [30–32]. It is worth noting that the success of most of the iterative methods above relies on the key step of using a spectral method to provide an initial guess that exhibits a correlation with the target signals. Typically, the initial estimate is the leading eigenvector of a data matrix, where the details of the definition are in (2) below.

1.2 Spectral Method

Our method comprises three steps. First, we construct a data matrix

$$\mathbf{D} \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n \mathcal{T}(y_i) \mathbf{a}_i \mathbf{a}_i^\top, \quad (2)$$

where $\mathcal{T} : \mathbb{R} \mapsto \mathbb{R}$ is a user-specified *preprocessing function*, with the purpose of mitigating the bias introduced by extreme observations y_i . Common choices of $\mathcal{T}$ can be found in [18, 20, 33, 34]. Second, we compute the r leading eigenvectors $\mathbf{x}_1, \dots, \mathbf{x}_r$ and the r trailing eigenvectors $\mathbf{x}_{d-r+1}, \dots, \mathbf{x}_d$ of $\mathbf{D}$. As will be demonstrated in Section 2, these eigenvectors reveal correlations with the target signals. In the final step, we form the optimal linear combinations of $\mathbf{x}_i$'s, by multiplying

$$\mathbf{X} \stackrel{\text{def}}{=} [\mathbf{x}_1, \dots, \mathbf{x}_r, \mathbf{x}_{d-r+1}, \dots, \mathbf{x}_d]$$

with a designed matrix $\mathbf{R}^*$, such that the correlation of the new estimator space and the target space is maximized. The resulting transformed subspace serves as the ultimate estimator.

While our primary focus is on the proportional regime in this paper, it is worth noting that there exists a simple intuitive understanding of the spectral method when $n \gg d$. From (2), $\mathbf{D}$ is the sample average of n i.i.d. rank-1 matrices $\mathcal{T}(y_i)\mathbf{a}_i\mathbf{a}_i^\top$. By the law of large numbers, we approximate $\mathbf{D}$ with the expectation matrix $\mathbb{E}[\mathcal{T}(y_i)\mathbf{a}_i\mathbf{a}_i^\top]$, which takes the form (see Appendix A)

$$c\mathbf{I}_d + \Xi C \Xi^\top,$$

for some $c \in \mathbb{R}$, $C \in \mathbb{R}^{r \times r}$ and $\Xi \stackrel{\text{def}}{=} [\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_r]$. The eigenvectors of this matrix have obvious correlations with our target signals.

To measure the performance of spectral methods on the subspace estimation, we employ the metric known as *principal angles* (see Section 2.2) between the target signals subspace $\text{span}\{\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_r\}$ and the estimation subspace generated using the eigenvectors of $\mathbf{D}$ associated with extreme eigenvalues. The principal angles can be viewed as a generalization of the angle between two vectors to the angles between two subspaces.

1.3 Principal Hessian Directions

Despite the prevalent application of the spectral initialization in the iterative algorithms above, there remains a gap in the literature regarding the comprehensive mathematical validation and analysis of the correlation between the initial estimators and target signals. The idea of the spectral method was initially inspired by principal Hessian directions [1] in the statistics literature designed for subspace estimation, with $\mathcal{T}(y_i) = y_i - \bar{y}$. In [1, 2], asymptotic performance analysis of principal Hessian directions was provided in the regime of $n \rightarrow \infty$ and fixed d, r . Later, the spectral method with $r = 1$ was introduced as an initialization step for the alternating minimization algorithm [35]. Most existing performance analyses of the spectral method focus on finite sample analysis [17, 18, 35]. Under Gaussian design, Chen and Candes [18] showed that the squared correlation between the eigenvector $\mathbf{x}_1$ and the target vector $\boldsymbol{\xi}$ approaches 1 with high probability when $n \geq c \cdot d$, where c is some sufficiently large constant. In a recent work [7], spectral methods with $r > 1$ were introduced for multiplexed phase retrieval with application to optical imaging in complex media, where phase transitions were discovered in the numerical results.

In [36], Lu and Li investigated the regime of $n, d \rightarrow \infty$ at a fixed ratio $\alpha = n/d$ for the $r = 1$ case and presented an exact analysis of the asymptotic performance of the spectral method. They showed, under Gaussian measurements, that the inner product of $\boldsymbol{\xi}$ and $\mathbf{x}_1$ converges in probability to a deterministic value depending on the observation model and sampling ratio α . Phase transitions characterization on the behavior of $\boldsymbol{\xi}^\top \mathbf{x}_1$ with respect to α was also provided. The above asymptotic analysis was initially formulated for the real-valued case and with the assumption that $\mathcal{T}(y)$ is non-negative [36]. Later in [37], Mondelli and Montanari generalized the asymptotic performance analysis to the complex-valued case and derived the results without the assumption that $\mathcal{T}(y) \geq 0$.

During the development of this paper, we noticed a recent work [38] on the two-signal mixed-GLM with a known link. Concurrently, [39] establishes optimal spectral thresholds for multi-index models, and [40] develops AMP-based spectral estimators with optimal preprocessing under correlated designs. Our focus differs: we allow unknown f , provide an asymptotic characterization for the rank- r spectral estimator for any finite $r \geq 1$, and reveal a phase transition in α , further justifying a refinement that also exploits trailing eigenvectors. It is important to note that the general rank- r spectral method studied in this paper introduces unique technical challenges that are not addressed in the previous works. Similar to [36] and [38], our performance analysis also reveals a phase transition phenomenon associated with the variations of the sampling ratio α . Moreover, we propose and justify a refinement of the spectral method that considers not

only the leading eigenvectors but also the trailing ones. We show that both sets of eigenvectors exhibit non-trivial correlations with the target signals.

1.4 Blessing of Dimensionality

Similar to the rank-1 spectral method in [36], stronger correlations would be achieved when the sampling ratio α increases. Nonetheless, unlike the previous study where only the leading eigenvector is considered, our findings reveal a notable distinction: we discover that both the leading eigenvectors and trailing eigenvectors capture the information of our signals. Another noteworthy remark is that the spectral method studied in [36] and this paper is characterized by its generality, as it does not require the prior knowledge of the model function $f(\cdot|\cdot)$. This advantage stems from the “blessing” of dimensionality (which contrasts with the conventional notion of the “curse” of dimensionality). The spectral methods extract information from the matrix $\mathbf{D}$ through its eigenvectors and form an estimator $\hat{\Xi}$. As we demonstrate in this paper, $\mathbf{x}_1, \dots, \mathbf{x}_r, \mathbf{x}_{d-r+1}, \dots, \mathbf{x}_d$ serve as the initial estimation of the input signals. Therefore, we could form a natural oracle estimator

$$\hat{\Xi} = \sum_{i \in \{1, \dots, r, d-r+1, \dots, d\}} w_i \mathbf{x}_i = \mathbf{X} \mathbf{R}^*$$

such that w_i , the entries of $\mathbf{R}^*$, are chosen to maximize the correlation with Ξ . As demonstrated in Theorem 3 below, using the principal angles as the metric, the optimal coefficients w_i that maximize the principal angles relative to the target subspace depend on $\Xi^\top \mathbf{X}$, which is generally not achievable in practice. Nonetheless, under the asymptotic regime, we are able to establish an approximation through their limit values. In Section 2, explicit formulas are provided for computing these limits.

1.5 Outline of the Paper

The rest of the paper is organized as follows. Our main results are stated in Section 2. First, we introduce assumptions and definitions, and then justify the assumption $\xi_i = \mathbf{e}_i$ (the i th canonical basis) without loss of generality. Then Theorem 1 provides the limits of the eigenvalues of $\mathbf{D}$ through the solutions of fixed-point equations. Following this, in Theorem 2 and Proposition 1, we characterize the limit of $\Xi^\top \mathbf{X}$, specifically the inner products of the extreme eigenvectors of $\mathbf{D}$ and the target signals (under the assumption $\xi_i = \mathbf{e}_i$, this is equivalent to the first r entries of the eigenvectors). In particular, Theorem 2 addresses the case where multiplicity occurs at the solutions of the fixed-point equations. Then due to the arbitrary choice of eigenbasis, it is not possible to identify the correlation between a pair of individual $\mathbf{x}_i$ and ξ_j . Instead, the aggregate correlation across the eigen-subspace attains convergence and is characterized. On the other hand, Proposition 1 discusses the special case without multiplicity, where the inner product for each pair of $\mathbf{x}_i$ and ξ_j attains a limit. In Theorem 3, we propose the postulate $\hat{\Xi} = \mathbf{X} \mathbf{R}$ as the estimator and construct the optimal $\mathbf{R}$ that minimizes the principal angles between the column spaces of $\hat{\Xi}$ and Ξ . To illustrate these results, we present worked examples with applications to multiplexed phase retrieval and the learning of two-layer neural networks in Section 3. Numerical simulations are provided there to verify the analytical results. The proofs of the main theorems are given in Section 4. Although we state and prove our results for the real-valued case, the analysis for the complex-valued case is essentially the same. See Remark 2 in Section 2 for how to adapt our results to the complex-valued problems. Finally, more technical details are included in the Appendix.

2 Main Results

2.1 General Assumptions and Definitions

As previously discussed, let $\xi_1, \dots, \xi_r \in \mathbb{R}^d$ be an orthonormal basis spanning a rank- r target subspace that we want to estimate, and let $\{\mathbf{a}_i \in \mathbb{R}^d\}_{1 \leq i \leq n}$ be the sensing vectors where $\mathbf{a}_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$. Given $\{\mathbf{a}_i^\top \xi_1, \dots, \mathbf{a}_i^\top \xi_r\}$, the i th measurement $y_i \in \mathbb{R}$ is drawn independently from

$$y_i \sim f(y | \mathbf{a}_i^\top \xi_1, \dots, \mathbf{a}_i^\top \xi_r), \quad 1 \leq i \leq n, \quad (3)$$

where $f(\cdot|\cdot)$ is a conditional density function modeling the sensing process. Additionally, we construct a data matrix

$$\mathbf{D} \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n \mathcal{T}(y_i) \mathbf{a}_i \mathbf{a}_i^\top, \quad (4)$$

where $\mathcal{T} : \mathbb{R} \mapsto \mathbb{R}$ is a user-specified preprocessing function, as discussed in the introduction. Moreover, define

$$\Xi \stackrel{\text{def}}{=} [\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_r] \in \mathbb{R}^{d \times r}, \quad \mathbf{X} \stackrel{\text{def}}{=} [\mathbf{x}_1, \dots, \mathbf{x}_r, \mathbf{x}_{d-r+1}, \dots, \mathbf{x}_d] \in \mathbb{R}^{d \times 2r} \quad (5)$$

where $\mathbf{x}_i$ is the eigenvector of $\mathbf{D}$ associated with the i -th eigenvalue in descending order. We examine this problem in the asymptotic regime where $n, d \rightarrow \infty$ with $n/d \rightarrow \alpha \in (0, \infty)$.

Before proceeding to further assumptions and definitions, we introduce an important simplification. Since the sensing vectors $\mathbf{a}_i$'s are drawn from the rotationally invariant multivariate normal distribution, any set of basis vectors $\{\boldsymbol{\xi}_1, \dots, \boldsymbol{\xi}_r\}$ for the target subspace is equivalent in the sense that the matrix $\Xi^\top \mathbf{X}$ retains the same probability distribution (see Appendix B for a proof). This allows us to assume

$$\boldsymbol{\xi}_i = \mathbf{e}_i, \quad 1 \leq i \leq r, \quad (6)$$

for the rest of this paper, where $\mathbf{e}_i$ is the i th canonical basis of $\mathbb{R}^d$.

As mentioned in the introduction, we use the principal angles defined below to measure the performance of our subspace estimation.

Definition 1 [41–43] Let $\mathcal{U}$ and $\mathcal{W}$ be two subspaces of $\mathbb{R}^d$, with dimensions p and q ($p < q$) respectively. Then the p **principal angles** $0 \leq \theta_1 \leq \dots \leq \theta_p \leq \pi/2$ between $\mathcal{U}$ and $\mathcal{W}$ are recursively defined for $k = 1, \dots, p$ by

$$\cos(\theta_k) = \max_{\mathbf{u} \in \mathcal{U}} \max_{\mathbf{w} \in \mathcal{W}} |\langle \mathbf{u}, \mathbf{w} \rangle| = |\langle \mathbf{u}_k, \mathbf{w}_k \rangle|$$

subject to the constraints $\|\mathbf{u}\| = \|\mathbf{w}\| = 1$, $\mathbf{u}^\top \mathbf{u}_j = 0$, $\mathbf{w}^\top \mathbf{w}_j = 0$ for $j = 1, \dots, k-1$. The vectors $\mathbf{u}_1, \dots, \mathbf{u}_p$ and $\mathbf{w}_1, \dots, \mathbf{w}_p$ are called the **principal vectors**.

In particular, the i -th principal angle between the column spaces of Ξ and $\mathbf{X}$ is given by $\cos^2 \theta_i = \lambda_i(\mathbf{X}^\top \Xi \Xi^\top \mathbf{X})$ for $i = 1, \dots, r$, where $\lambda_i(\cdot)$ is the i -th largest eigenvalue (see Theorem 1 of [41]). In Section 2.2, we will provide exact asymptotic analysis for the matrix $\Xi^\top \mathbf{X}$, from which we can identify the limiting behavior of the principal angles.

Definition 2 Define random vector $\mathbf{S} \stackrel{\text{def}}{=} [S_1, \dots, S_r]^\top$ along with the random variables $\tilde{Y}$ and Y such that $Y = \mathcal{T}(\tilde{Y})$ and

$$\begin{aligned} S_1, \dots, S_r &\stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1), \\ \tilde{Y} | S_1, \dots, S_r &\sim f(\tilde{y} | s_1, \dots, s_r) \end{aligned} \quad (7)$$

where $f(\cdot|\cdot)$ is the conditional density function in (3) representing the observation model. We assume that $\mathbb{P}(Y = 0) < 1$ and Y has bounded support. Let τ_u be the supremum and τ_l be the infimum of this support, i.e.

$$\tau_u = \inf\{a : \mathbb{P}(Y \leq a) = 1\}, \quad \tau_l = \sup\{a : \mathbb{P}(Y \geq a) = 1\}. \quad (8)$$

To lighten the notations, we define several auxiliary functions below which will play significant roles in our later technical discussions.

Definition 3 Define

$$\psi_\alpha(t) \stackrel{\text{def}}{=} \frac{t}{\alpha} + \mathbb{E} \left[\frac{tY}{t - Y} \right], \quad (9)$$

over the intervals $t \in (-\infty, \tau_l) \cup (\tau_u, \infty)$. One can verify that $\psi_\alpha(t)$ is strictly convex over the interval (τ_u, ∞) and strictly concave over the interval $(-\infty, \tau_l)$. Thus there exists a unique minimum of $\psi_\alpha(t)$ over (τ_u, ∞) and a unique maximum of $\psi_\alpha(t)$ over $(-\infty, \tau_l)$. Denote

$$\bar{t}_u = \underset{t > \tau_u}{\operatorname{argmin}} \psi_\alpha(t) \quad \text{and} \quad \bar{t}_l = \underset{t < \tau_l}{\operatorname{argmax}} \psi_\alpha(t). \quad (10)$$

Consider a modified version of $\psi_\alpha(t)$:

$$\zeta_\alpha(t) \stackrel{\text{def}}{=} \begin{cases} \psi_\alpha(\max\{t, \bar{t}_u\}), & \text{if } t \in (\tau_u, \infty), \\ \psi_\alpha(\min\{t, \bar{t}_l\}), & \text{if } t \in (-\infty, \tau_l). \end{cases} \quad (11)$$

This function $\zeta_\alpha(t)$ is non-decreasing over $(-\infty, \tau_l)$ and (τ_u, ∞) .

Definition 4 Define $\mathbf{M}_t^* \stackrel{\text{def}}{=} \mathbb{E} \left[\frac{tY}{t-Y} \mathbf{S}\mathbf{S}^\top \right]$ over $t \in (-\infty, \tau_l) \cup (\tau_u, \infty)$ and let $h_i(t)$ be the i -th eigenvalue of $\mathbf{M}_t^*$, for $i = 1, \dots, r$. Note that over the interval $t \in (-\infty, \bar{t}_l) \cup (\bar{t}_u, \infty)$, ψ_α attains an inverse function. This allows us to change the variable from t to λ , where λ is defined by $\lambda = \psi_\alpha(t)$. Consequently, $\mathbf{M}_t^*$ can be re-expressed as a function of λ , denoted by $\mathbf{M}^*(\lambda)$. Switching between the two variables t and λ would make it more convenient for our subsequent analysis.

Definition 5 (Stochastic domination [44]) Given two families of nonnegative random variables

$$Z = \{Z^{(n)} : n \in \mathbb{N}\}, \quad W = \{W^{(n)} : n \in \mathbb{N}\},$$

we say Z is **stochastically dominated** by W (denoted by $Z \prec W$) if for all (small) $\epsilon > 0$ and (large) $D > 0$, we have

$$\mathbb{P}(Z^{(n)} > n^\epsilon W^{(n)}) \leq n^{-D}$$

for sufficiently large n . If $|Z| \prec W$, we write $Z = O_\prec(W)$.

Assumption 1 For the eigenvalue of $\mathbf{M}_t^*$, assume

$$\lim_{t \rightarrow \tau_l} h'_i(t) = -\infty \quad \text{and} \quad \lim_{t \rightarrow \tau_u} h'_i(t) = -\infty, \quad i = 1, \dots, r.$$

Note an immediate consequence is that $\lim_{t \rightarrow \tau_l} h_i(t) = \infty$ and $\lim_{t \rightarrow \tau_u} h_i(t) = \infty$. By the definition of $\mathbf{M}_t^*$, this assumption essentially guarantees Y has a sufficient mass near the boundaries τ_l and τ_u .

2.2 Main Results

Under the assumptions and definitions in Section 2.1, we present the following results characterizing the eigenvalues and eigenvectors of $\mathbf{D}$.

Theorem 1 The following results characterize the eigenvalues of $\mathbf{D}$:

1. For any $\alpha > 0$ and for each $i = 1, \dots, r$, there exists a solution to the equation

$$\zeta_\alpha(t) = h_i(t), \quad t > \tau_u, \quad (12)$$

Denote the r solutions as $t_{1,u}^*, \dots, t_{r,u}^*$ (not necessarily distinct) in non-ascending order. Similarly, for any $\alpha > 0$ and for each $i = 1, \dots, r$, there exists a solution to the equation

$$\zeta_\alpha(t) = h_i(t), \quad t < \tau_l. \quad (13)$$

Denote the r solutions as $t_{1,l}^*, \dots, t_{r,l}^*$ (not necessarily distinct) in non-descending order.

2. For the largest eigenvalues of $\mathbf{D}$, we have

$$\lambda_i(\mathbf{D}) = \zeta_\alpha(t_{i,u}^*) + O_\prec \left(\frac{1}{\sqrt{d}} \right), \quad i = 1, \dots, r \quad (14)$$

and for the smallest eigenvalues of $\mathbf{D}$, we have

$$\lambda_{d-i+1}(\mathbf{D}) = \zeta_\alpha(t_{i,l}^*) + O_\prec \left(\frac{1}{\sqrt{d}} \right), \quad i = 1, \dots, r. \quad (15)$$

Our simplification (6) implies that $\mathbf{x}_i^\top \boldsymbol{\xi}_j$ equals the j -th entry of $\mathbf{x}_i$. With the notations from Theorem 1 above, we describe the asymptotic behavior of the inner products $\mathbf{x}_i^\top \boldsymbol{\xi}_j$ in the following.

Theorem 2 For any $t^* \in \{t_{1,u}^*, \dots, t_{r,u}^*\} \cup \{t_{1,l}^*, \dots, t_{r,l}^*\}$, $\lambda^* = \zeta_\alpha(t^*)$ is an eigenvalue of $\mathbf{M}_{t^*}^*$. Suppose λ^* has multiplicity k and $\mathbf{V} \stackrel{\text{def}}{=} [\mathbf{v}_1, \dots, \mathbf{v}_k]$ where columns form an arbitrary eigenbasis associated with λ^* . Then there exist eigenvalues $\{\tilde{\lambda}_{i_1}, \dots, \tilde{\lambda}_{i_k}\}$ (assume no multiplicity) of $\mathbf{D}$ converging to λ^* as $d \rightarrow \infty$. Let the associated eigenvectors compose $\tilde{\mathbf{X}} \stackrel{\text{def}}{=} [\mathbf{x}_{i_1}, \dots, \mathbf{x}_{i_k}]$. Then

$$\Xi^\top \tilde{\mathbf{X}} \tilde{\mathbf{X}}^\top \Xi \xrightarrow{\mathcal{P}} \mathbb{1}_{\zeta_{\alpha'}(t^*) > 0} \cdot \mathbf{V} \left(\mathbf{I} + \mathbf{V}^\top \frac{\mathbf{E}(t^*)}{\zeta_{\alpha'}(t^*)} \mathbf{V} \right)^{-1} \mathbf{V}^\top \quad (16)$$

where $\mathbb{1}_{\{\cdot\}}$ is the indicator function and $\mathbf{E}(t) \stackrel{\text{def}}{=} \mathbb{E} \left[\frac{Y^2}{(t-Y)^2} \mathbf{S} \mathbf{S}^\top \right]$.

Proposition 1 Consider the λ^* defined in Theorem 2 but assume it has no multiplicity. Let λ^* be the i -th eigenvalue of $\mathbf{M}_{t^*}^*$ with the associated eigenvector $\mathbf{v}$. Let $\mathbf{x}$ be the eigenvector of $\mathbf{D}$ associated with the eigenvalue converging to λ^* . Then

$$\Xi^\top \mathbf{x} \xrightarrow{\mathcal{P}} \pm \sqrt{\frac{\mathbb{1}_{\zeta_{\alpha'}(t^*) > 0}}{1 - h'_i(t^*)/\zeta_{\alpha'}(t^*)}} \mathbf{v} \quad (17)$$

where the $\pm$ sign is due to the sign ambiguity of the eigenvectors $\mathbf{x}, \mathbf{v}$.

Recall the definition of $\mathbf{X}$ given in (5). From Theorem 2, we know that $\mathbf{X}$ exhibits a non-trivial correlation with our target Ξ . Therefore, we postulate our estimator as

$$\hat{\Xi} = \mathbf{X} \mathbf{R} \quad (18)$$

where $\mathbf{R} \in \mathbb{R}^{2r \times r}$ satisfies $\mathbf{R}^\top \mathbf{R} = \mathbf{I}_r$. Furthermore, in the case where the low-dimensional matrix $\mathbf{M}_{t^*}^*$ exhibits no multiplicity, it is not only possible to determine the limit of $\Xi^\top \mathbf{X} \mathbf{X}^\top \Xi$ but also to obtain a precise inner product $\Xi^\top \mathbf{X}$. Under the setting of Proposition 1 (no multiplicity), our next theorem provides the optimal $\mathbf{R}$ minimizing the principal angles between the column spaces of Ξ and $\hat{\Xi}$.

Theorem 3 (Optimal Estimator) Define $\mathbf{R}^* \in \mathbb{R}^{2r \times r}$ such that its columns comprise the leading r eigenvectors of $\mathbf{X}^\top \Xi \Xi^\top \mathbf{X}$. Consider $\theta_1, \dots, \theta_r$ as the principal angles between the column spaces of Ξ and $\hat{\Xi}$. Then $\mathbf{R}^*$ solves the optimization problem

$$\max_{\substack{\mathbf{R} \in \mathbb{R}^{2r \times r} \\ \text{s.t. } \mathbf{R}^\top \mathbf{R} = \mathbf{I}_r}} \sum_{i=1}^r \cos(\theta_i). \quad (19)$$

Remark 1 (Phase transition) The theorems above provide a complete solution to the exact analysis of the multi-rank spectral methods in high dimension. From Theorem 1 and Theorem 2, we discover generic phase transitions depending on the sign of $\zeta_{\alpha'}(t^*)$. When the derivative is positive, the extreme eigenvector of $\mathbf{D}$ has a nonzero projection onto the target subspace and thus contains information about the target signals. Otherwise, all the correlations between the extreme eigenvectors and the r target signals vanish.

Remark 2 (The complex-valued case) The results of Theorem 1 and Theorem 2 are applicable to the complex-valued case with the following modifications: (1) For the observation model, we now assume $\boldsymbol{\xi}_i \in \mathbb{C}^d$, the sensing vectors $\mathbf{a}_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(\mathbf{0}, \mathbf{I}_d)$ and the sensing model in (3) is now $y_i \sim f(y | |\langle \mathbf{a}, \boldsymbol{\xi}_1 \rangle|, \dots, |\langle \mathbf{a}, \boldsymbol{\xi}_r \rangle|)$. (2) Correspondingly, in Definition 2, the random variable S_i is now drawn independently from $\mathcal{CN}(0, 1)$ instead of $\mathcal{N}(0, 1)$ and we have $\tilde{Y} | |S_1|, \dots, |S_r| \sim f(\tilde{y} | |s_1|, \dots, |s_r|)$

Remark 3 (The optimal preprocessing functions) The matrix $\mathbf{D}$ in (4) is constructed as an empirical average of rank-1 matrices, weighted by the observations. To avoid the strong influence of extreme observations, a preprocessing function $\mathcal{T}$ is often applied to observations y_i . Examples of commonly used preprocessing functions include trimming in [18] and the subset indicator function in [20]. Furthermore, [34] studied the optimal design of $\mathcal{T}$. In our case, the assumption in (8) guarantees that after applying the preprocessing function, Y has bounded support in (τ_l, τ_u) , in order to avoid outlier observations.

3 Examples

3.1 Multiplexed Phase Retrieval

To show how our results can be applied in practice, we first present an example corresponding to the multiplexed phase retrieval model [7]. In particular, we consider the following sensing model

$$y_i = \sum_{j=1}^r \beta_j (\mathbf{a}_i^\top \boldsymbol{\xi}_j)^2, \quad (20)$$

where $\beta_1, \dots, \beta_r \in \mathbb{R}$ are the source strength parameters of the corresponding signals. We use the subset preprocessing function

$$\mathcal{T}(y) = \mathbb{1}_{\{y \geq a\}}. \quad (21)$$

This model appears in real-life scenarios where the physical detector acquires measurements from several indistinguishable source signals. Imaging in complex media such as biological tissues would be one such example [7].

Particularly, one can check that $\mathbf{M}_{t^*}^*$ is diagonal in this example. Therefore, the eigenvectors are simply $\mathbf{e}_j$ from the canonical basis and thus the vector $\Xi^\top \mathbf{x}$ in Proposition 1 only has one nonzero entry. This implies that each $\mathbf{x}$ is correlated with exactly one signal vector $\boldsymbol{\xi}_j$. Furthermore, we can simplify the expression of $h_i(\lambda)$ and $\psi_\alpha(\lambda)$ as

$$h_i(t) = \frac{C_i t}{t-1} \quad \text{and} \quad \psi_\alpha(t) = t \left(1/\alpha + \frac{D}{t-1} \right),$$

where

$$C_i \stackrel{\text{def}}{=} \mathbb{E}[Y S_i^2] \quad \text{and} \quad D \stackrel{\text{def}}{=} \mathbb{E}[Y].$$

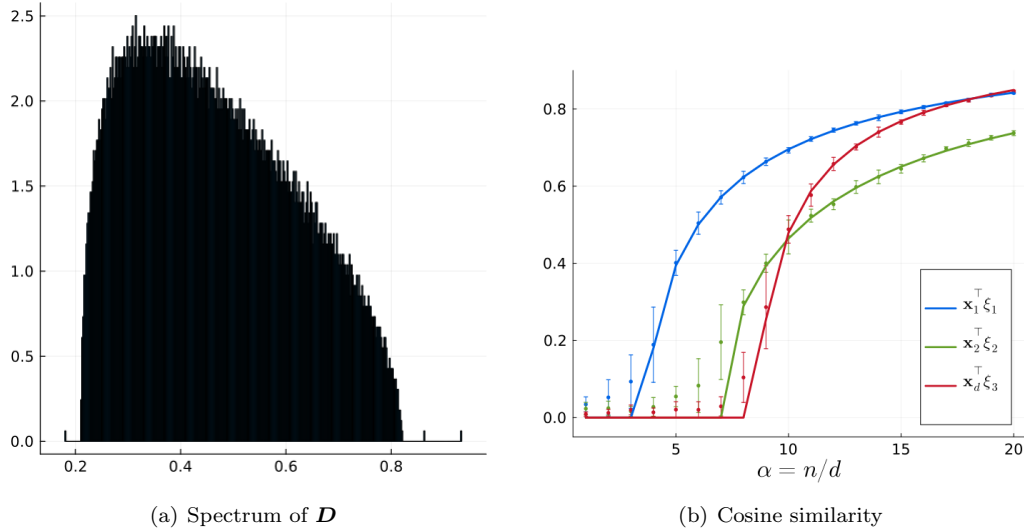


Figure 1: (a) Histogram for the eigenvalues of $\mathbf{D}$ with $d = 8192$ and $\alpha = 20$ under the multiplexed phase retrieval model. (b) Analytical predictions and numerical simulations. Numerical results (for $d = 8192$) are averaged over 10 independent trials, with the error bars indicating ± 1 standard deviation.

Figure 1 shows the spectrum of $\mathbf{D}$ and the analytical predictions of the performance of our spectral method, compared with numerical simulation results. Three extreme eigenvalues (called *spikes*) are clear in the histogram. In particular, we have two spikes corresponding to the largest eigenvalues and one spike corresponding to the smallest eigenvalue. In our experiments, the number of signals $r = 3$ and the source strength parameters are set to $\beta_1 = 2$, $\beta_2 = 1$ and $\beta_3 = -1.5$. The threshold in the subset preprocessing function is set to $a = 1.2$. The figure demonstrates that our analytical predictions match numerical results.

3.2 Shallow Neural Networks

Now we show a more general example where the $(\mathbf{x}_i, \boldsymbol{\xi}_j)$ pairs are not decoupled to be one-to-one corresponding. We apply Theorem 2 to get the asymptotic characterization of the correlations between extreme eigenvectors and the target signals. Consider the learning of a two-layer neural network, where the model is formulated as

$$y_i = \sum_{j=1}^r \gamma_j \max(0, \mathbf{a}_i^\top \boldsymbol{\xi}_j). \quad (22)$$

The $\max(0, \cdot)$ is the rectified linear unit (ReLU) function and γ_j 's are the weights in the second layer of the network. In this experiment, we also consider the subset preprocessing function defined in (21). Then we get

$$\mathbf{M}_t^* = \frac{t\mathbf{C}}{t-1} \quad \text{and} \quad \psi_\alpha(t) = t \left(1/\alpha + \frac{D}{t-1} \right),$$

where

$$\mathbf{C} \stackrel{\text{def}}{=} \mathbb{E}[\mathbf{Y}\mathbf{S}\mathbf{S}^\top] \in \mathbb{R}^{r \times r} \quad \text{and} \quad D \stackrel{\text{def}}{=} \mathbb{E}[Y].$$

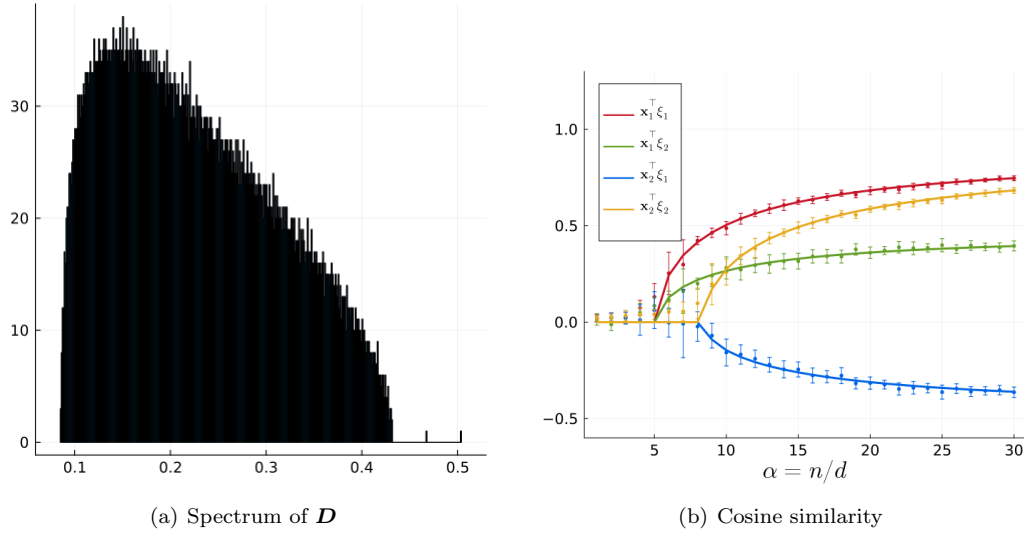


Figure 2: (a) Histogram for the eigenvalue of $\mathbf{D}$ with $d = 8192$ and $\alpha = 30$ under the shallow neural network model. (b) Analytical predictions and numerical simulations for the cosine similarity. Numerical results (for $d = 8192$) are averaged over 10 independent trials, with the error bars indicating ± 1 standard deviation.

Figure 2 shows the spectrum of $\mathbf{D}$ and the analytical predictions and simulations of the performance of our estimators. In this example, $r = 2$ and the weights are set to $\gamma_1 = 1.2$ and $\gamma_2 = 1$. The threshold for the preprocessing function is set to $a = 1.5$. From the spectrum of $\mathbf{D}$, we identify only two spikes corresponding to the largest eigenvalues. Moreover, in this example, each pair $(\mathbf{x}_i, \boldsymbol{\xi}_j)$ has a nonzero correlation. Again, the figure demonstrates a clear alignment between analytical predictions and numerical results, affirming that applying our theoretical results above yields accurate predictions of the principal angles between the eigenvectors subspace and the target subspace.

In the previous examples, $\mathbf{M}_{t^*}^*$ only has simple eigenvalues. Although rare, it is possible to encounter scenarios where some eigenvalues of $\mathbf{M}_{t^*}^*$ have multiplicity. In this case, any eigen-basis could be generated to represent the eigen-subspace. It becomes impossible to characterize $\Xi^\top \mathbf{x}$ for each $\mathbf{x}$. Nevertheless, the formula in Theorem 2 remains applicable to derive the limit of $\Xi^\top \tilde{\mathbf{X}} \tilde{\mathbf{X}}^\top \Xi$, where columns of $\tilde{\mathbf{X}}$ are formed by any eigen-basis corresponding to the eigenvalue with multiplicity.

3.3 Example with eigenvalue multiplicity

We present such a special example here for $r = 2$. Consider a designed model

$$y_i = \delta_1(\mathbf{a}_i^\top \boldsymbol{\xi}_j) + \delta_2(\mathbf{a}_i^\top \boldsymbol{\xi}_j)^2 \quad (23)$$

with the truncation preprocessing function

$$\mathcal{T}(y) = \begin{cases} \tau_u & \text{if } y > \tau_u \\ y & \\ \tau_l & \text{if } y < \tau_l \end{cases} \quad (24)$$

Specifically, we set $\delta_1 = \frac{37}{45}$, $\delta_2 = -\frac{3}{25}$, $\tau_u = \frac{283}{90}$, $\tau_l = -3$ and $\alpha = 18.54380488623075$. Figure 3 below plots the spectrum of $\mathbf{D}$ and we can see one spike from the largest eigenvalue and two spikes from the smallest eigenvalues. The two trailing spikes are extremely close in the simulation and one can verify by our analytic results that they will be exactly equal when $n, d \rightarrow \infty$.

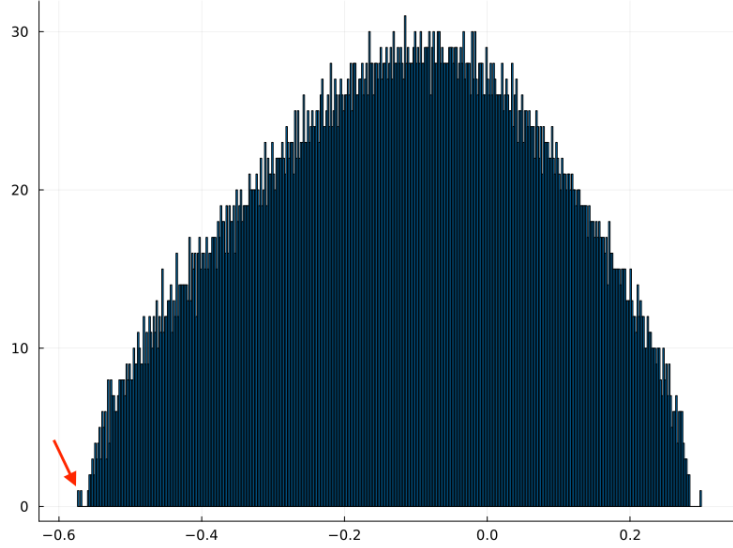


Figure 3: Histogram for the eigenvalue of $\mathbf{D}$ with $d = 8192$ under the multiplicity model. The plot reveals one leading spike and two closely adjacent trailing spikes, as indicated by the arrow.

Now $\Xi^\top \mathbf{x}_1$ converges according to Proposition 1, with analytic predictions provided by (17). Conversely, the two vectors $\Xi^\top \mathbf{x}_d$ and $\Xi^\top \mathbf{x}_{d-1}$ do not have limits, as illustrated in Figure 4(a). In this case, (16) in Theorem 2 can be utilized to characterize $\Xi^\top \tilde{\mathbf{X}} \tilde{\mathbf{X}}^\top \Xi$, where $\tilde{\mathbf{X}} = [\mathbf{x}_d \ \mathbf{x}_{d-1}] \in \mathbb{R}^{d \times 2}$ (see Figure 4(b)).

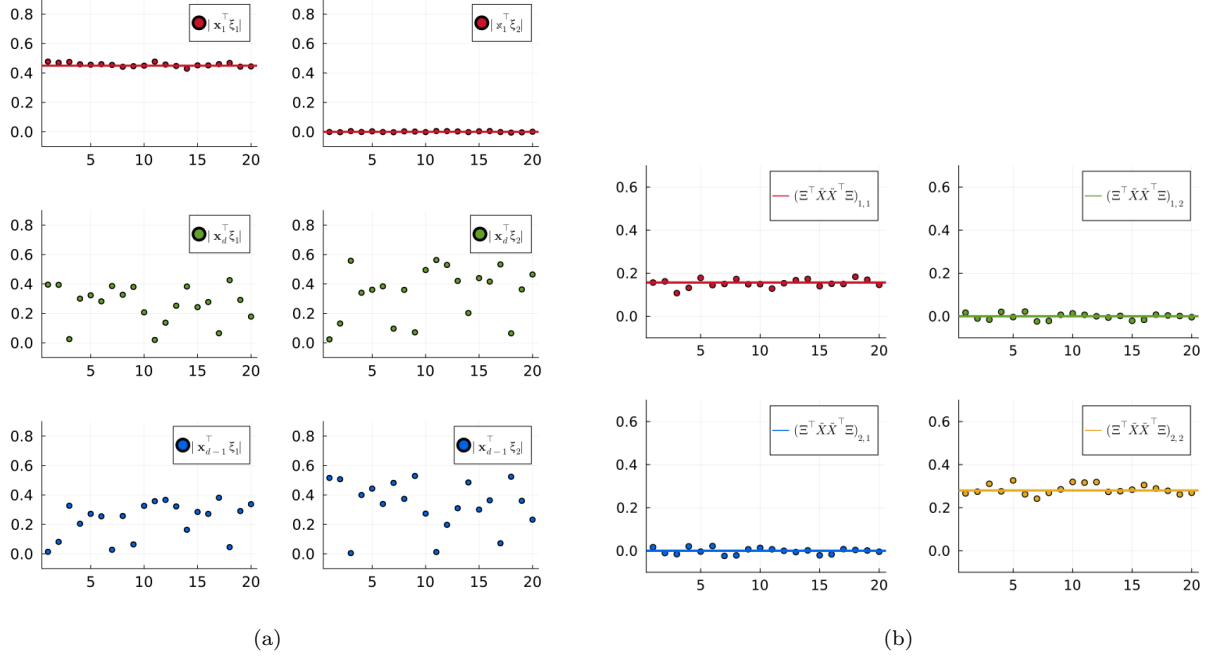


Figure 4: We set $d = 8192$ and run the simulation over 20 trials. (a) Simulation for entries of $\Xi^\top \mathbf{x}_1$, $\Xi^\top \mathbf{x}_d$ and $\Xi^\top \mathbf{x}_{d-1}$. Horizontal lines represent the analytic prediction for $\Xi^\top \mathbf{x}_1$ by (17). (b) Simulation for entries of $\Xi^\top \tilde{\mathbf{X}} \tilde{\mathbf{X}}^\top \Xi$. Horizontal lines represent the analytic predictions by (16).

4 Proof of Main Results

Write the data matrix $\mathbf{D}$ defined in (4) in a block-partitioned form as

$$\mathbf{D} = \begin{bmatrix} \mathbf{A} & \mathbf{q}^\top \\ \mathbf{q} & \mathbf{P} \end{bmatrix}, \quad (25)$$

where $\mathbf{A} \in \mathbb{R}^{r \times r}$, $\mathbf{q} \in \mathbb{R}^{(d-r) \times r}$ and $\mathbf{P} \in \mathbb{R}^{(d-r) \times (d-r)}$. Then partition the $\mathbf{a}_i$ in (4) as $\mathbf{a}_i = \begin{bmatrix} \mathbf{s}_i \\ \mathbf{u}_i \end{bmatrix}$, where $\mathbf{s}_i \in \mathbb{R}^r$ and $\mathbf{u}_i \in \mathbb{R}^{d-r}$. Construct $\mathbf{S} \in \mathbb{R}^{r \times n}$ and $\mathbf{U} \in \mathbb{R}^{(d-r) \times n}$ such that their i -th columns are $\mathbf{s}_i$ and $\mathbf{u}_i$, respectively. Moreover, for simplicity, in the proof of our main results, we shall omit the preprocessing function $\mathcal{T}(\cdot)$ from the definition of $\mathbf{D}$ in (2) (effectively treating $\mathcal{T}(\cdot)$ as the identity map). Hence, Equation (2) simplifies to

$$\mathbf{D} = \frac{1}{n} \sum_{i=1}^n y_i \mathbf{a}_i \mathbf{a}_i^\top.$$

If we define $\mathbf{Y} \stackrel{\text{def}}{=} \text{diag}(y_1, y_2, \dots, y_n)$, then

$$\begin{aligned} \mathbf{A} &= \frac{1}{n} \sum_{i=1}^n y_i \mathbf{s}_i \mathbf{s}_i^\top = \frac{1}{n} \mathbf{S} \mathbf{Y} \mathbf{S}^\top \\ \mathbf{q} &= \frac{1}{n} \sum_{i=1}^n y_i \mathbf{u}_i \mathbf{s}_i^\top = \frac{1}{n} \mathbf{U} \mathbf{Y} \mathbf{S}^\top \\ \mathbf{P} &= \frac{1}{n} \sum_{i=1}^n y_i \mathbf{u}_i \mathbf{u}_i^\top = \frac{1}{n} \mathbf{U} \mathbf{Y} \mathbf{U}^\top. \end{aligned}$$

4.1 Fixed-Point Characterization of the Eigenvalues

For any complex number z outside the spectrum of $\mathbf{P}$, define

$$\mathbf{M}(z) \stackrel{\text{def}}{=} \mathbf{A} + \mathbf{q}^\top (z\mathbf{I} - \mathbf{P})^{-1} \mathbf{q} \quad (26)$$

and the i -th eigenvalue of $\mathbf{M}(z)$ as $\tilde{h}_i(z)$. The subsequent lemma establishes a connection between the extreme eigenvalues of $\mathbf{D}$ and the eigenvalues of the matrices defined in (26).

Lemma 1 *For any $\tilde{\lambda}$ outside the spectrum of $\mathbf{P}$, $\tilde{\lambda}$ is an eigenvalue of $\mathbf{D}$ if and only if there exists an eigenvalue $\tilde{h}_i(\tilde{\lambda})$ of $\mathbf{M}(\tilde{\lambda})$ that has the same multiplicity k and satisfy*

$$\tilde{\lambda} = \tilde{h}_i(\tilde{\lambda}) \quad (27)$$

proof: For any $\tilde{\lambda}$ not an eigenvalue of $\mathbf{P}$, apply Schur's determinant identity to get

$$\begin{aligned} \det(\tilde{\lambda}\mathbf{I} - \mathbf{D}) &= \det(\tilde{\lambda}\mathbf{I} - \mathbf{P}) \det(\tilde{\lambda}\mathbf{I} - \mathbf{A} - \mathbf{q}^\top (\tilde{\lambda}\mathbf{I} - \mathbf{P})^{-1} \mathbf{q}) \\ &= \det(\tilde{\lambda}\mathbf{I} - \mathbf{P}) \det(\tilde{\lambda}\mathbf{I} - \mathbf{M}(\tilde{\lambda})). \end{aligned}$$

Therefore $\det(\tilde{\lambda}\mathbf{I} - \mathbf{D}) = 0$ is equivalent to $\det(\tilde{\lambda}\mathbf{I} - \mathbf{M}(\tilde{\lambda})) = 0$, which implies the fixed-point equations that $\tilde{\lambda}$ is equal to the eigenvalues of $\mathbf{M}(\tilde{\lambda})$. $\blacksquare$

Lemma 1 offers an approach to study the eigenvalues of $\mathbf{D}$ that lie outside the spectrum of $\mathbf{P}$, which we call *spikes*, through the eigenvalues of $\mathbf{M}(\tilde{\lambda})$. In the following, Lemma 2 introduces a fix-point equation that provides another way to characterize these eigenvalues. The combination of these two equations is instrumental in the subsequent analysis of the spikes. Thereafter, we characterize the eigenvalues of $\mathbf{M}(\tilde{\lambda})$ and their limits in Lemma 3. Proofs of Lemmas 2 and 3 are provided in Appendix C.

Lemma 2 *For any eigenvalue of $\tilde{\lambda}$ of $\mathbf{D}$ outside the spectrum of $\mathbf{P}$, we have*

$$\tilde{\lambda} = \psi_\alpha(\tilde{t}) + O_{\prec} \left(\frac{1}{\sqrt{d}} \right) \quad (28)$$

where $\tilde{t} \stackrel{\text{def}}{=} \frac{\alpha}{m(\tilde{\lambda})}$ and $m(\tilde{\lambda}) \stackrel{\text{def}}{=} \frac{1}{d-r} \text{Tr}(\tilde{\lambda}\mathbf{I} - \mathbf{P})^{-1}$.

Lemma 3 *For any $z \in \mathbb{C}$ outside the spectrum of $\mathbf{P}$, let $m^*(z)$ be the unique solution to:*

$$z = \frac{1}{\alpha} \cdot \frac{\alpha}{m^*(z)} + \frac{\alpha}{m^*(z)} \cdot \mathbb{E} \left[\frac{Y}{\frac{\alpha}{m^*(z)} - Y} \right]$$

and $t^* = \frac{\alpha}{m^*(z)}$. Then $t^* \in \{t_{1,u}^*, \dots, t_{r,u}^*\} \cup \{t_{1,l}^*, \dots, t_{r,l}^*\}$ and we have

$$\|\mathbf{M}(z) - \mathbf{M}_{t^*}^*\| = O_{\prec} \left(\frac{1}{\sqrt{d}} \right). \quad (29)$$

The deviation of the i -th eigenvalues of $\mathbf{M}(z)$ and $\mathbf{M}_{t^*}^*$ (denoted $\tilde{h}_i(t^*)$ and $h_i(t^*)$, respectively) also have the same order:

$$\tilde{h}_i(t^*) = h_i(t^*) + O_{\prec} \left(\frac{1}{\sqrt{d}} \right) \quad (30)$$

Note that in our subsequent proofs for concentration results, instead of using the original definition of stochastic domination, the more convenient Theorem 7.7 of [44] is often applied:

Lemma 4 (Theorem 7.7 of [44]) *For $n \in \mathbb{N}$, let $\{X_i^{(n)}\}_{1 \leq i \leq n}$ and $\{Y_i^{(n)}\}_{1 \leq i \leq n}$ be independent families of random variables and $\{a_{ij}^{(n)}\}_{1 \leq i, j \leq n}$ and $\{b_i^{(n)}\}_{1 \leq i \leq n}$ be deterministic values. Suppose that all entries $X_i^{(n)}$ and $Y_i^{(n)}$ are independent and satisfy*

$$\mathbb{E}X = 0, \quad \mathbb{E}|X|^2 = 1, \quad \|X\|_p \stackrel{\text{def}}{=} (\mathbb{E}|X|^p)^{1/p} \leq \mu_p$$

for all $p \in \mathbb{N}$ and some constant μ_p . Then we have the bounds

$$\begin{aligned}\sum_i b_i X_i &\prec \left(\sum_i |b_i|^2 \right)^{1/2} \\ \sum_{i,j} a_i X_i Y_j &\prec \left(\sum_{i,j} |a_{ij}|^2 \right)^{1/2} \\ \sum_{i \neq j} a_i X_i X_j &\prec \left(\sum_{i \neq j} |a_{ij}|^2 \right)^{1/2}.\end{aligned}$$

4.2 Proof of Theorem 1

We prove (12) and (14) since the arguments are analogous for (13) and (15). First, to verify the existence of the solution to (12), recall that Assumption 1 ensures $\lim_{t \rightarrow \tau_u} h_i(t) = \infty$. Moreover, as $t \rightarrow \infty$, $h_i(t)$ converges to the i -th eigenvalue of $\mathbb{E}[Y \mathbf{S} \mathbf{S}^\top]$ since $\mathbf{M}_t^* = \mathbb{E}[Y \mathbf{S} \mathbf{S}^\top] + \mathbb{E}[\frac{Y^2}{t-Y} \mathbf{S} \mathbf{S}^\top]$, while $\zeta_\alpha(t)$ goes to infinity. Therefore, the existence of the solutions is evident by the fact that $\zeta_\alpha(t)$ and $h_i(t)$ must intersect.

We now proceed to show (14). We observe that $\mathbf{P}$ forms a principal submatrix of $\mathbf{D}$. The relationship between their eigenvalues is governed by the Cauchy Interlacing Theorem [45, Theorem 4.3.28], which implies the inequalities $\lambda_{k+r}(\mathbf{D}) \leq \lambda_k(\mathbf{P})$ and $\lambda_k(\mathbf{P}) \leq \lambda_k(\mathbf{D})$ for $k = 1, \dots, d-r$. By applying the first inequality with index $k = i$ and the second with index $k = i+r$, we construct a sandwich for $\lambda_{i+r}(\mathbf{D})$:

$$\lambda_{i+r}(\mathbf{P}) \leq \lambda_{i+r}(\mathbf{D}) \leq \lambda_i(\mathbf{P}).$$

Define $a \stackrel{\text{def}}{=} \lambda_{\min}(\mathbf{P})$ and $b \stackrel{\text{def}}{=} \lambda_{\max}(\mathbf{P})$. Let a^* and b^* be their respective limits, which represent the boundaries of the limiting spectral distribution (LSD) of $\mathbf{P}$. Notably, the LSD of $\mathbf{P}$ is a generalized version of the Marchenko-Pastur law and is extensively studied in existing literature (for example, see [46] or [47]). In particular,

$$b^* = \psi_\alpha(\bar{t}_u) \quad \text{and} \quad \lambda_i(\mathbf{P}) = b^* + O_{\prec} \left(\frac{1}{\sqrt{d}} \right)$$

The concentration of the extreme eigenvalues of $\mathbf{P}$ at the spectral edge b^* is a standard result from random matrix theory on eigenvalue rigidity [48]. Therefore, for a fixed $i \in \{1, 2, \dots, d-r\}$, by interlacing, we have established

$$\lambda_{i+r}(\mathbf{D}) = \psi_\alpha(\bar{t}_u) + O_{\prec} \left(\frac{1}{\sqrt{d}} \right)$$

Leveraging Lemmas 1 and 2, we are able to characterize any spike $\tilde{\lambda}$ of $\mathbf{D}$ ($\tilde{\lambda} > b$) through the fixed-point equation

$$\tilde{\lambda} = \tilde{h}_i(\tilde{t}) = \psi_\alpha(\tilde{t}) + O_{\prec} \left(\frac{1}{\sqrt{d}} \right) \tag{31}$$

where $\tilde{t} = \frac{\alpha}{m(\tilde{\lambda})}$ and $m(\tilde{\lambda}) = \frac{1}{d-r} \text{Tr}(\tilde{\lambda} \mathbf{I} - \mathbf{P})^{-1}$. A simple analysis of $m(\lambda)$ would reveal its monotonicity for any $\lambda > b$. Therefore, variables λ and t have one-to-one correspondence. To facilitate further discussion, we introduce the notation $\tilde{H}_i(\lambda) \stackrel{\text{def}}{=} \tilde{h}_i(t(\lambda))$, effectively reinterpreting $\tilde{h}_i$ as a function of λ , and denote $\tilde{\lambda}_i$ as the solution that satisfies

$$\tilde{\lambda}_i = \tilde{H}_i(\tilde{\lambda}_i). \tag{32}$$

Note that $\tilde{H}_i(\tilde{\lambda}_i)$'s are recognized as the eigenvalues of $\mathbf{M}(\tilde{\lambda}_i)$ and (32) is exactly the same fix-point equation described in Lemma 1. In the following, we categorize the spikes $\tilde{\lambda}_i$ based on their limiting behavior. Certain spikes, referred to as *hidden spikes*, will be shown to have b^* as their limit, while others, designated as *salient spikes*, converge to a value larger than b^* . More precisely, for $i = 1, \dots, r$, we establish that

$$\tilde{\lambda}_i = \begin{cases} \psi_\alpha(t_{i,u}^*) + O_{\prec} \left(\frac{1}{\sqrt{d}} \right), & \text{if } \psi'(t_{i,u}^*) \geq 0 \\ b^* + O_{\prec} \left(\frac{1}{\sqrt{d}} \right), & \text{otherwise.} \end{cases} \tag{33}$$

Upon validating (33), the statement (14) in Theorem 1 follows by applying the relation $b^* = \psi_\alpha(\bar{t}_u)$ and the definition of ζ_α .

To prove (33), it suffices to connect equation (32) to its corresponding limiting version. Specifically, we aim to bound the difference between $\tilde{\lambda}_i$ and $\lambda^* \stackrel{\text{def}}{=} \psi_\alpha(t^*)$, where t^* is defined as the solution to

$$h_i(t^*) = \psi_\alpha(t^*) \quad (34)$$

Our analysis proceeds by considering two separate cases below.

Case I: $\lambda^* > b^*$. Assume λ^* has multiplicity k as an eigenvalue of $\mathbf{M}_{t^*}^*$. For the simplicity of the proof, let $\lambda^* = h_1^*(t^*) = h_2^*(t^*) = \dots = h_k^*(t^*)$ without loss of generality. Note that $\mathbf{M}(\tilde{\lambda}_i)$ exhibits no multiplicity with high probability [49]. We will show that k spikes $\tilde{\lambda}_i = \tilde{h}_i(\tilde{t}_i), i = 1, \dots, k$ have the same limit $\psi_\alpha(t^*)$. The following discussion pertains to a fixed i .

Analogous to $\tilde{H}_i(\lambda)$, but using the inverse of ψ instead (restricted to the segment with $\lambda^* > b^* = \psi_\alpha(\bar{t}_u)$), we can also reinterpret h_i as a function of the variable λ and define $H_i(\lambda) \stackrel{\text{def}}{=} h_i(t(\lambda))$. Consequently, equation (34) implies that λ^* satisfies

$$\lambda^* = H_i(\lambda^*). \quad (35)$$

Define $g(x) \stackrel{\text{def}}{=} \tilde{H}_i(x) - x$, by the monotonicity of $\tilde{H}_i$, one can verify that $|g'(x)| \geq 1$ for any x (since $\tilde{H}_i'(x) < 0$). Furthermore, Lemma 3 implies

$$g(\tilde{\lambda}_i) - g(\lambda^*) = \lambda^* - \tilde{H}_i(\lambda^*) = \lambda^* - H_i(\lambda^*) + O_{\prec} \left(\frac{1}{\sqrt{d}} \right).$$

Recalling the bound of $g'(\cdot)$, we utilize the mean value theorem to derive $|\tilde{\lambda}_i - \lambda^*| = O_{\prec} \left(\frac{1}{\sqrt{d}} \right)$ and therefore $\tilde{\lambda}_i$ converges to a limit $\lambda^* = \psi_\alpha(t^*)$.

Case II: $\lambda^* \leq b^*$. Recall that λ^* is the solution to (35) and b^* is the limit of $b \stackrel{\text{def}}{=} \lambda_{\max}(\mathbf{P})$. Consider a fixed small $\epsilon > 0$. The monotonicity property yields

$$H_i(b + \epsilon) < b + \epsilon$$

and as a result, for a sufficiently large dimension d , we have

$$\tilde{H}_i(b + \epsilon) < b + \epsilon.$$

Since $\tilde{H}_i$ is also monotone, it can be inferred that all solutions $\tilde{\lambda}_i$ to (32) are bounded above by $b + \epsilon$ for sufficiently large d . Therefore, $\tilde{\lambda}_i$ will be hidden in the bulk ($\tilde{\lambda}_i \leq b$) and we have shown using interlacing that $\tilde{\lambda}_i \rightarrow b^*$. This substantiates our verification of (33) and thus completes the proof of Theorem 1. ■

4.3 Characterization of the Eigenvectors

Recall that the term *spikes* refers to the eigenvalues of $\mathbf{D}$ that fall outside the spectrum of $\mathbf{P}$. Lemma 5 below explores the correlations between the target signals and the eigenvectors of $\mathbf{D}$ corresponding to its spikes, where the eigenvectors of $\mathbf{M}(\tilde{\lambda})$ are involved in the characterization. Consequently, the focus transitions to examining the asymptotic properties of the $r \times r$ matrix $\mathbf{M}(\tilde{\lambda})$. This analysis is detailed in Lemmas 6 and 7, where Cauchy's Residue Theorem plays a crucial role.

Lemma 5 *For any spike $\tilde{\lambda}$ of $\mathbf{D}$, consider $\mathbf{x}$ as an associated eigenvector and define $\bar{\mathbf{x}} \stackrel{\text{def}}{=} \Xi^\top \mathbf{x}$. It follows that $\bar{\mathbf{x}}$ is a scaled eigenvector of $\mathbf{M}(\tilde{\lambda})$ corresponding with eigenvalue $\tilde{\lambda}$ and satisfies*

$$\|\bar{\mathbf{x}}\|^2 + \bar{\mathbf{x}}^\top \mathbf{q}^\top (\tilde{\lambda} \mathbf{I} - \mathbf{P})^{-2} \mathbf{q} \bar{\mathbf{x}} = 1. \quad (36)$$

Specifically, when $\tilde{\lambda}$ has multiplicity 1 and $\mathbf{v}$ denotes the associated unit-norm eigenvector of $\mathbf{M}(\tilde{\lambda})$, then

$$\bar{\mathbf{x}} \bar{\mathbf{x}}^\top = \frac{\mathbf{v} \mathbf{v}^\top}{1 + \mathbf{v}^\top \mathbf{q}^\top (\tilde{\lambda} \mathbf{I} - \mathbf{P})^{-2} \mathbf{q} \mathbf{v}} \quad (37)$$

proof: Note that an immediate consequence of the assumption (6) is that $\bar{\mathbf{x}}$ is the first r entries of $\mathbf{x}$. Write $\mathbf{D}$ as the block-partitioned form in (25) and consider

$$\mathbf{x} = \begin{bmatrix} \bar{\mathbf{x}} \\ \bar{\mathbf{y}} \end{bmatrix}.$$

The relation $\mathbf{D}\mathbf{x} = \tilde{\lambda}\mathbf{x}$ leads to

$$\begin{cases} \mathbf{A}\bar{\mathbf{x}} + \mathbf{q}^\top \bar{\mathbf{y}} = \tilde{\lambda}\bar{\mathbf{x}} \\ \mathbf{q}\bar{\mathbf{x}} + \mathbf{P}\bar{\mathbf{y}} = \tilde{\lambda}\bar{\mathbf{y}}. \end{cases}$$

Given that $\tilde{\lambda}$ is a spike, the matrix $\tilde{\lambda}\mathbf{I} - \mathbf{P}$ is invertible. Simplifying the above system yields

$$\bar{\mathbf{y}} = (\tilde{\lambda}\mathbf{I} - \mathbf{P})^{-1}\mathbf{q}\bar{\mathbf{x}}$$

and subsequently

$$\mathbf{A}\bar{\mathbf{x}} + \mathbf{q}^\top (\tilde{\lambda}\mathbf{I} - \mathbf{P})^{-1}\mathbf{q}\bar{\mathbf{x}} = \tilde{\lambda}\bar{\mathbf{x}}. \quad (38)$$

By Lemma 1, $\tilde{\lambda}$ is affirmed as an eigenvalue of $\mathbf{M}(\tilde{\lambda}) \stackrel{\text{def}}{=} \mathbf{A} + \mathbf{q}^\top (\tilde{\lambda}\mathbf{I} - \mathbf{P})^{-1}\mathbf{q}$, and it follows that $\bar{\mathbf{x}}$ is an eigenvector of $\mathbf{M}(\tilde{\lambda})$ corresponding to $\tilde{\lambda}$. Given our convention that the eigenvectors have unit norms, $\|\mathbf{x}\|^2 = \|\bar{\mathbf{x}}\|^2 + \|\bar{\mathbf{y}}\|^2 = 1$, leading to equation (36). Equation (37) naturally arises from scaling $\mathbf{v}$ by the magnitude of $\bar{\mathbf{x}}$. ■

Lemma 6 *For any eigenvalue of $\tilde{\lambda}$ of $\mathbf{D}$ with multiplicity k , let $\{\mathbf{x}_{i_1}, \dots, \mathbf{x}_{i_k}\}$ be any basis of the associated eigensubspace and define $\bar{\mathbf{x}}_{i_j} \stackrel{\text{def}}{=} \Xi^\top \mathbf{x}_{i_j}$ for $j = 1, \dots, k$. Suppose γ is a simple closed contour in $\mathbb{C}$ that exclusively encircles the eigenvalue $\tilde{\lambda}$, then we have*

$$\sum_{j=1}^k \bar{\mathbf{x}}_{i_j} \bar{\mathbf{x}}_{i_j}^\top = \frac{1}{2\pi\iota} \oint_{\gamma} (z\mathbf{I} - \mathbf{M}(z))^{-1} dz \quad (39)$$

proof: Lemma 9 in Appendix D confirms that $(z\mathbf{I} - \mathbf{M}(z))^{-1}$ is well-defined and analytic within the region enclosed by γ except at λ . For any $z \in \mathbb{C}$ outside the spectrum of $\mathbf{D}$, the resolvent of $\mathbf{D}$ is defined as $\mathbf{G}(z) = (z\mathbf{I} - \mathbf{D})^{-1} = \sum_{i=1}^d \frac{\mathbf{x}_i \mathbf{x}_i^\top}{z - \lambda_i}$, where each $(\lambda_i, \mathbf{x}_i)$ denotes an eigenpair of $\mathbf{D}$. By applying Cauchy's Integral Formula, one can derive that $\sum_{j=1}^k \mathbf{x}_{i_j} \mathbf{x}_{i_j}^\top = \frac{1}{2\pi\iota} \oint_{\gamma} \mathbf{G}(z) dz$ (for example, see [50]). As a result, we have

$$\sum_{j=1}^k \bar{\mathbf{x}}_{i_j} \bar{\mathbf{x}}_{i_j}^\top = \Xi^\top \left(\frac{1}{2\pi\iota} \oint_{\gamma} \mathbf{G}(z) dz \right) \Xi \quad (40)$$

$$= \frac{1}{2\pi\iota} \oint_{\gamma} \left(z\mathbf{I} - \mathbf{A} - \mathbf{q}^\top (z\mathbf{I} - \mathbf{P})^{-1}\mathbf{q} \right)^{-1} dz \quad (41)$$

$$= \frac{1}{2\pi\iota} \oint_{\gamma} (z\mathbf{I} - \mathbf{M}(z))^{-1} dz. \quad (42)$$

where the second equality follows from the application of the inverse matrix formula for 2×2 block matrix (for example, see [45]) on $\mathbf{D}$. ■

The preceding lemma demonstrated that the correlation between the eigenvectors and target signal vectors can be calculated through an integration over the contour that encircles the corresponding eigenvalue $\tilde{\lambda}$. For the efficient computation of this contour integral, the application of Cauchy's Residue Theorem is instrumental, transforming the task into the simple calculation of residues.

allowing us to simplify the problem into the calculation of residues. The explicit formula for this calculation is articulated in Lemma 7 below. The detailed proof of this lemma is available in Appendix D.

Lemma 7 *Fix any $t^* \in \{t_{1,u}^*, \dots, t_{r,u}^*\} \cup \{t_{1,l}^*, \dots, t_{r,l}^*\}$ as specified in Theorem 1 and define $\lambda^* \stackrel{\text{def}}{=} \psi_\alpha(t^*)$. According to Theorem 1 and the definition of h_i , λ^* is an eigenvalue of $\mathbf{M}_{t^*}^*$. Assuming this eigenvalue*

has multiplicity k , define the matrix $\mathbf{V} \stackrel{\text{def}}{=} [\mathbf{v}_1, \dots, \mathbf{v}_k]$ whose columns form a basis of the corresponding eigensubspace. Given $\psi'(t^*) > 0$, we have

$$\lim_{z \rightarrow \lambda^*} (z - \lambda^*)(z\mathbf{I} - \mathbf{M}^*(z))^{-1} = \mathbf{V} \left(\mathbf{I} + \mathbf{V}^\top \frac{\mathbf{E}(t^*)}{\psi'(t^*)} \mathbf{V} \right)^{-1} \mathbf{V}^\top \quad (43)$$

where $\mathbf{M}^*(\cdot)$ is defined in Definition 4 and $\mathbf{E}(t) \stackrel{\text{def}}{=} \mathbb{E} \left[\frac{Y^2}{(t-Y)^2} \mathbf{S} \mathbf{S}^\top \right]$.

4.4 Proof of Theorem 2

The proof is presented for the case $t^* \in \{t_{1,u}^*, \dots, t_{r,u}^*\}$, and analogous reasoning applies to the other case. For any $\tilde{\lambda} \in \{\tilde{\lambda}_{i_1}, \dots, \tilde{\lambda}_{i_k}\}$, Theorem 1 outlines two potential limits for $\tilde{\lambda}$: either $\tilde{\lambda} \rightarrow b^*$ with $b^* = \zeta_\alpha(\bar{t}_u)$, or $\tilde{\lambda} \rightarrow \zeta_\alpha(t^*) > b^*$, where t^* is the solution to equation (12).

Case I: $\lambda^* = b^*$. In this scenario, $t^* \leq \bar{t}_u$. Our analysis will primarily focus on $\tilde{\lambda}_{i_1}$, though the argument extends similarly to other eigenvalues. Again, we focus on the situation where the eigenvalue $\tilde{\lambda}_{i_1}$ exhibits no multiplicity [49]. Lemma 5 established the relation:

$$\bar{\mathbf{x}}_{i_1} \bar{\mathbf{x}}_{i_1}^\top = \frac{\mathbf{v}_1 \mathbf{v}_1^\top}{1 + \mathbf{v}_1^\top \mathbf{q}^\top (\tilde{\lambda}_{i_1} \mathbf{I} - \mathbf{P})^{-2} \mathbf{q} \mathbf{v}_1} \quad (44)$$

As $\tilde{\lambda}_{i_1}$ approaches b^* , consider a fixed small $\epsilon > 0$, ensuring for a sufficiently large d , $\tilde{\lambda}_{i_1} < b^* + \epsilon$. By monotonicity, it follows that

$$1 + \mathbf{v}_1^\top \mathbf{q}^\top (\tilde{\lambda}_{i_1} \mathbf{I} - \mathbf{P})^{-2} \mathbf{q} \mathbf{v}_1 > 1 + \mathbf{v}_1^\top \mathbf{q}^\top ((b^* + \epsilon) \mathbf{I} - \mathbf{P})^{-2} \mathbf{q} \mathbf{v}_1.$$

Note that $\mathbf{M}'(b^* + \epsilon) = -\mathbf{q}^\top ((b^* + \epsilon) \mathbf{I} - \mathbf{P})^{-2} \mathbf{q}$. Combining Assumption 1 and the connection between $\mathbf{M}(\cdot)$ and $\mathbf{M}^*(\cdot)$ in Lemma 3, we have $\|\mathbf{M}'(b^* + \epsilon)\| \rightarrow \infty$. Therefore, $\bar{\mathbf{x}}_{i_1} \bar{\mathbf{x}}_{i_1}^\top \rightarrow \mathbf{0}$ and eventually we have $\sum_{j=1}^k \bar{\mathbf{x}}_{i_j} \bar{\mathbf{x}}_{i_j}^\top \rightarrow \mathbf{0}$.

Case II: $\lambda^* > b^*$. As assumed in Theorem 1, $\tilde{\lambda}_{i_1}, \dots, \tilde{\lambda}_{i_k}$ are simple eigenvalues and therefore distinct. Define γ as a simple closed contour in $\mathbb{C}$ that encloses $\tilde{\lambda}_{i_1}, \dots, \tilde{\lambda}_{i_k}$ excluding other eigenvalues. By Lemma 6, we have

$$\sum_{j=1}^k \bar{\mathbf{x}}_{i_j} \bar{\mathbf{x}}_{i_j}^\top = \frac{1}{2\pi i} \oint_{\gamma} (z\mathbf{I} - \mathbf{M}(z))^{-1} dz \quad (45)$$

As dimension $d \rightarrow \infty$, we impose the constraint that γ only encircles the eigenvalues $\tilde{\lambda}_{i_1}, \dots, \tilde{\lambda}_{i_k}$ of matrix $\mathbf{D}$. As $\mathbf{M}(z) \rightarrow \mathbf{M}_{t^*}^*$ uniformly for the corresponding t^* defined in Lemma 3. Moreover, $z\mathbf{I} - \mathbf{M}(z)$ is always non-singular along the contour γ (since $\tilde{\lambda}_{i_1}, \dots, \tilde{\lambda}_{i_k}$ are precisely the solutions to $z\mathbf{I} - \mathbf{M}(z)$ and we guaranteed that z does not coincide with any of them). Therefore,

$$\frac{1}{2\pi i} \oint_{\gamma} (z\mathbf{I} - \mathbf{M}(z))^{-1} dz \rightarrow \frac{1}{2\pi i} \oint_{\gamma^*} (z\mathbf{I} - \mathbf{M}^*(z))^{-1} dz \quad (46)$$

where γ^* is a simple closed contour that only encircles λ^* and no other eigenvalues of $\mathbf{M}^*(\lambda^*)$. Leveraging the Cauchy Residue Theorem, the evaluation of the integral on the right is reduced to computing the residue of $(z\mathbf{I} - \mathbf{M}^*(z))^{-1}$ at λ^* , as detailed in Lemma 7, thus concluding the proof of Theorem 2. ■

4.5 Proof of Proposition 1

The foundational work of the proof has already been laid out in the general case presented in Theorem 2. For Proposition 1, we specifically address the scenario where the multiplicity $k = 1$. In this context, the right side of (16) degenerates to the case of rank-1 matrices, given $\mathbf{V} = \mathbf{v}_1 \in \mathbb{R}^{r \times 1}$. To establish Equation (17), it suffices to show that $\mathbf{v}_1^\top \mathbf{E}(t^*) \mathbf{v}_1 = -h_i'(t^*)$.

Let $\mathbf{M}_t^* = \mathbf{A}^* + \mathbf{K}(t)$, where $\mathbf{A}^* \stackrel{\text{def}}{=} \mathbb{E}[Y \mathbf{S} \mathbf{S}^\top]$ and $\mathbf{K}(t) \stackrel{\text{def}}{=} \mathbb{E}[\frac{Y^2}{t-Y} \mathbf{S} \mathbf{S}^\top]$. Taking the derivative with respect to t in the equation

$$(\mathbf{A}^* + \mathbf{K}(t)) \mathbf{v}_1 = h_i(t) \mathbf{v}_1.$$

yields

$$-E(t)\mathbf{v}_1 + (\mathbf{A}^* + \mathbf{K}(t))\mathbf{v}_1'(t) = h_i'(t)\mathbf{v}_1 + h_i(t)\mathbf{v}_1'(t).$$

Considering the orthogonality $\mathbf{v}_1(t)^\top \mathbf{v}_1'(t) = 0$ (due to the unit-norm convention on the eigenvector $\mathbf{v}_1$), we multiply $\mathbf{v}_1^\top$ on the left and simplify the equation to get

$$\mathbf{v}_1^\top E(t)\mathbf{v}_1 = -h_i'(t).$$

This argument completes the proof of Proposition 1. ■

4.6 Proof of Theorem 3

Denote $\mathbf{J} = \mathbf{X}^\top \Xi \Xi^\top \mathbf{X}$ and consider its eigendecomposition

$$\mathbf{J} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top,$$

where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_{2r})$ and $\mathbf{Q} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{2r}]$. Given Ξ and $\hat{\Xi}$ both have orthonormal columns, the cosines of principal angles are given by the singular values of $\hat{\Xi}^\top \Xi$ (see Theorem 1 in [41], for example). Consequently, the optimization problem in (19) is equivalent to

$$\max_{\substack{\mathbf{R} \in \mathbb{R}^{2r \times r} \\ \text{s.t. } \mathbf{R}^\top \mathbf{R} = \mathbf{I}_r}} \text{Tr}(\mathbf{R}^\top \mathbf{J} \mathbf{R}) \quad (47)$$

Denote the columns of $\mathbf{Q}^\top \mathbf{R}$ as $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_r$. Considering the eigendecomposition of $\mathbf{J}$, the optimization reduces to

$$\max_{\substack{\mathbf{R} \in \mathbb{R}^{2r \times r} \\ \text{s.t. } \mathbf{R}^\top \mathbf{R} = \mathbf{I}_r}} \sum_{i=1}^r \sum_{j=1}^{2r} \lambda_j \mathbf{c}_{i,j}^2 = \max_{\substack{\mathbf{R} \in \mathbb{R}^{2r \times r} \\ \text{s.t. } \mathbf{R}^\top \mathbf{R} = \mathbf{I}_r}} \sum_{j=1}^{2r} \lambda_j \left(\sum_{i=1}^r \mathbf{c}_{i,j}^2 \right) = \max_{\substack{\mathbf{R} \in \mathbb{R}^{2r \times r} \\ \text{s.t. } \mathbf{R}^\top \mathbf{R} = \mathbf{I}_r}} \sum_{j=1}^{2r} \lambda_j a_j,$$

if we denote $a_j \stackrel{\text{def}}{=} \sum_{i=1}^r \mathbf{c}_{i,j}^2$ and $\mathbf{c}_{i,j}$ as the j -th entry of $\mathbf{c}_i$. One can verify that each $a_j \leq 1$ and the sum $\sum_{j=1}^{2r} a_j = r$. With $\lambda_1 \geq \dots \geq \lambda_{2r}$, the maximum of our optimization problem is attained when $a_1 = \dots = a_r = 1$ and $a_{r+1} = \dots = a_{2r} = 0$, achievable by setting the first r rows of $\mathbf{Q}^\top \mathbf{R}^*$ to be $\mathbf{I}_r$ and the remaining r rows to be zero. Equivalently, $\mathbf{R}^*$ comprises the first r columns of $\mathbf{Q}$ and therefore the leading r eigenvectors of $\mathbf{J}$.

The only caveat here is that $\mathbf{X}$ has sign ambiguity for its column vectors. In other words, any $\mathbf{X}\mathbf{B}$ would be a valid new $\mathbf{X}$ for a diagonal matrix $\mathbf{B} \in \mathbb{R}^{2r \times 2r}$ with $B_{ii} \in \{\pm 1\}$. This is equivalent to multiplying $\mathbf{R}$ by $\mathbf{B}$ on the left. Nonetheless, our process to derive $\mathbf{R}^*$ from $\mathbf{Q}$ demonstrates that this sign ambiguity does not affect the outcome, as $\mathbf{Q}$ itself is composed of eigenvectors that inherently possess sign ambiguity. Thus, we conclude the proof of Theorem 3, which elucidates the construction of the optimal estimator $\hat{\Xi}$ based on the initial estimator $\mathbf{X}$. ■

A Intuition for the Spectral Method

In the formulation presented by (2), $\mathbf{D}$ is the sample average of n i.i.d. rank-1 matrices $\mathcal{T}(y_i) \mathbf{a}_i \mathbf{a}_i^\top$, where each $y_i \sim f(y | \mathbf{a}_i^\top \Xi)$ for $i = 1, 2, \dots, n$ and $\Xi = [\boldsymbol{\xi}_1, \boldsymbol{\xi}_2, \dots, \boldsymbol{\xi}_r]$. We introduce the orthogonal matrix $\mathbf{Q} = [\Xi \Xi^\perp] \in \mathbb{R}^{d \times d}$, comprising the matrix Ξ in its first r columns and the orthogonal complement of Ξ in the subsequent $d - r$ column. Define $\tilde{\mathbf{a}}_i \stackrel{\text{def}}{=} \mathbf{Q}^\top \mathbf{a}_i$ and $c_i \stackrel{\text{def}}{=} \mathcal{T}(y_i)$. Note that c_i depends solely on the first r components of $\tilde{\mathbf{a}}_i$, which leads to the block-partitioned form:

$$\tilde{\mathbf{a}}_i = \begin{bmatrix} \tilde{\mathbf{s}}_i \\ \tilde{\mathbf{u}}_i \end{bmatrix},$$

where $\tilde{\mathbf{s}}_i \in \mathbb{R}^r$ and $\tilde{\mathbf{u}}_i \in \mathbb{R}^{d-r}$. Rewriting the data matrix in (2), we get

$$\mathbf{D} \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n c_i \mathbf{a}_i \mathbf{a}_i^\top = \mathbf{Q} \left[\frac{1}{n} \sum_{i=1}^n c_i \tilde{\mathbf{a}}_i \tilde{\mathbf{a}}_i^\top \right] \mathbf{Q}^\top.$$

When n is large, by the law of large numbers,

$$\begin{aligned}
\mathbf{D} &\approx \mathbf{Q} \mathbb{E}[c_i \tilde{\mathbf{a}}_i \tilde{\mathbf{a}}_i^\top] \mathbf{Q}^\top \\
&= \begin{bmatrix} \Xi & \Xi^\perp \end{bmatrix} \begin{bmatrix} \mathbb{E}[c_i \tilde{\mathbf{s}}_i \tilde{\mathbf{s}}_i^\top] & \mathbf{0} \\ \mathbf{0} & \mathbb{E}(c_i) \mathbf{I}_{d-r} \end{bmatrix} \begin{bmatrix} \Xi^\top \\ (\Xi^\perp)^\top \end{bmatrix} \\
&= \Xi \left[\mathbb{E}[c_i \tilde{\mathbf{s}}_i \tilde{\mathbf{s}}_i^\top] - \mathbb{E}(c_i) \mathbf{I}_r \right] \Xi^\top + \mathbb{E}(c_i) \mathbf{I}_d.
\end{aligned}$$

This derivation indicates that $\mathbf{D}$ can be viewed as a rank- r perturbation of a scaled identity matrix. Particularly, if the eigenvalues of $\mathbb{E}[c_i \tilde{\mathbf{s}}_i \tilde{\mathbf{s}}_i^\top]$ are distinct from $\mathbb{E}(c_i)$, then the eigenvectors of $\mathbf{D}$ associated with its extreme eigenvalues effectively recover the signal space. This provides an intuitive argument for signal recovery via the spectral method. Nonetheless, it is important to note the difference between this intuition and the spectral method presented in this paper, as we analyze under the regime where both d and n grow to infinity proportionally.

B Justification of the Assumption $\xi_i = \mathbf{e}_i$

In this appendix, we show that without loss of generality, we can assume that $\xi_i = \mathbf{e}_i$ for $i = 1, \dots, r$. First, define $\Xi = [\xi_1, \dots, \xi_r] \in \mathbb{R}^{d \times r}$ and $\mathbf{E}_r = [\mathbf{e}_1, \dots, \mathbf{e}_r]$ (the canonical basis vectors). It is always possible to find an orthogonal matrix $\mathbf{Q}$ such that $\Xi = \mathbf{Q} \mathbf{E}_r$. Consider $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ as the matrix composed of sensing vectors. The outputs of the spectral method are based on $\mathbf{X} \stackrel{\text{def}}{=} [\mathbf{x}_1, \xi_2, \dots, \mathbf{x}_r, \mathbf{x}_{d-r+1}, \dots, \mathbf{x}_d]$, the eigenvectors of

$$\mathbf{D} = \mathbf{A} \text{diag}(\mathcal{T}(\mathbf{a}_1^\top \Xi), \dots, \mathcal{T}(\mathbf{a}_n^\top \Xi)) \mathbf{A}^\top,$$

By defining $\mathbf{a}'_i = \mathbf{Q}^\top \mathbf{a}_i$ and $\mathbf{A}' = [\mathbf{a}'_1, \dots, \mathbf{a}'_n]$, we can reformulate the data matrix $\mathbf{D}$ as

$$\mathbf{D} = \mathbf{Q} \mathbf{A}' \text{diag}(\mathcal{T}(\mathbf{a}'_1^\top \mathbf{E}_r), \dots, \mathcal{T}(\mathbf{a}'_n^\top \mathbf{E}_r)) \mathbf{A}'^\top \mathbf{Q}^\top.$$

Due to the rotational invariance of i.i.d. Gaussian distributions, $\mathbf{A}'$ follows the same distribution as $\mathbf{A}$. Consequently, the distribution of $\mathbf{X}$ with Ξ as the target is equivalent to the distribution of $\mathbf{Q} \mathbf{X}$ with the new target $\mathbf{E}_r$. Moreover, the correlations between the eigenvectors and the target signals are the same under any orthogonal transform of target signals. $\blacksquare$

C Lemmas for Eigenvalue Characterizations

Lemma 8 For any $z \in \mathbb{C}$ with $\Im(z) > 0$, define the resolvent and the Stieltjes transform as

$$\mathbf{G}(z) \stackrel{\text{def}}{=} (\mathbf{P} - z\mathbf{I})^{-1} \quad \text{and} \quad s(z) \stackrel{\text{def}}{=} \frac{1}{d-r} \text{Tr} \mathbf{G}(z),$$

respectively. For a fixed $i \in \{1, \dots, d-r\}$, define

$$\mathbf{G}^{(i)} \stackrel{\text{def}}{=} \left(\frac{1}{n} \sum_{j \neq i}^n y_j \mathbf{u}_j \mathbf{u}_j^\top - z\mathbf{I} \right)^{-1} \quad \text{and} \quad s^{(i)}(z) \stackrel{\text{def}}{=} \frac{1}{d-r} \mathbf{u}_i^\top \mathbf{G}^{(i)} \mathbf{u}_i.$$

Then the relationship between $s(z)$ and $s^{(i)}(z)$ is given by:

$$s(z) = s^{(i)}(z) + O_{\prec} \left(\frac{1}{\sqrt{d}} \right) \quad (48)$$

proof: Recall $\mathbf{P} \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i \mathbf{u}_i \mathbf{u}_i^\top = \frac{1}{n} \mathbf{U} \mathbf{Y} \mathbf{U}^\top$. Denote $\mathbf{P}^{(i)} \stackrel{\text{def}}{=} \frac{1}{n} \sum_{j \neq i}^n \mathbf{y}_j \mathbf{u}_j \mathbf{u}_j^\top$, $E \stackrel{\text{def}}{=} \Re(z)$ and $\eta \stackrel{\text{def}}{=} \Im(z)$. Applying Woodbury's matrix inverse lemma [51], we get

$$\begin{aligned} \mathbf{G} &= (\mathbf{P} - z\mathbf{I})^{-1} \\ &= \left(\mathbf{P}^{(i)} - z\mathbf{I} + \frac{1}{n} \mathbf{y}_i \mathbf{u}_i \mathbf{u}_i^\top \right)^{-1} \\ &= \mathbf{G}^{(i)} - \frac{\frac{\mathbf{y}_i}{n} \mathbf{G}^{(i)} \mathbf{u}_i \mathbf{u}_i^\top \mathbf{G}^{(i)}}{1 + \frac{\mathbf{y}_i}{n} \mathbf{u}_i^\top \mathbf{G}^{(i)} \mathbf{u}_i} \end{aligned}$$

Therefore, $\mathbf{G}$ is a rank-1 perturbation of $\mathbf{G}^{(i)}$. Let $F(x) \stackrel{\text{def}}{=} \frac{1}{d-r} \#\{\lambda_k(\mathbf{P}) \leq x\}$ and $F^{(i)}(x) \stackrel{\text{def}}{=} \frac{1}{d-r} \#\{\lambda_k(\mathbf{P}^{(i)}) \leq x\}$. Through eigenvalue interlacing, we deduce $\sup_x (d-r)|F(x) - F^{(i)}(x)| \leq 1$. Using integration by parts yields

$$\begin{aligned} \left| s(z) - \frac{1}{d-r} \text{Tr} \mathbf{G}^{(i)} \right| &= \left| \int \frac{1}{x-z} dF(x) - \int \frac{1}{x-z} dF^{(i)}(x) \right| \\ &= \left| \int \frac{F(x)}{(x-z)^2} d(x) - \int \frac{F^{(i)}(x)}{(x-z)^2} d(x) \right| \\ &= \frac{1}{d-r} \left| \int \frac{(d-r)F(x) - (d-r)F^{(i)}(x)}{(x-E-\eta)^2} d(x) \right| \\ &\leq \frac{1}{d-r} \int \frac{1}{(x+E)^2 + \eta^2} dx \\ &\leq \frac{1}{d-r} \frac{C}{\eta} \\ &= O_{\prec} \left(\frac{1}{d} \right). \end{aligned} \tag{49}$$

for $\eta > 0$. The extension of this identity from z in the complex upper half-plane to the real axis relies on the uniform convergence of the Stieltjes transform. As established in works like [52], the convergence $m(z) \rightarrow m^*(z)$ holds uniformly for z in compact subsets of $\mathbb{C}^+$, and the limiting transform is continuous down to the real axis, justifying taking the limit $\Im(z) \rightarrow 0$.

Furthermore, the limiting spectrum of $\mathbf{P}$ is a generalized version of the Marchenko-Pastur law and can be shown to be bounded [46]. Combined with the assumption that the spectrum of $\mathbf{Y}$ has a bounded support, we ascertain that

$$\begin{aligned} &\left| s^{(i)}(z) - \frac{1}{d-r} \text{Tr} \mathbf{G}^{(i)} \right| \\ &= \left| \frac{1}{d-r} \mathbf{u}_i^\top \mathbf{G}^{(i)} \mathbf{u}_i - \frac{1}{d-r} \text{Tr} \mathbf{G}^{(i)} \right| \\ &= \left| \frac{1}{d-r} \mathbf{u}_i^\top \text{diag}(\lambda_1(\mathbf{G}^{(i)}), \dots, \lambda_{d-r}(\mathbf{G}^{(i)})) \mathbf{u}_i - \frac{1}{d-r} \text{Tr} \mathbf{G}^{(i)} \right| \text{ (rotational invariance of Gaussians)} \\ &= \left| \frac{1}{d-r} \sum_{j=1}^{d-r} \lambda_j(\mathbf{G}^{(i)}) (g_j^2 - 1) \right| \quad \text{(each } g_j \sim \mathcal{N}(0, 1) \text{ is the } j\text{-th entry of } \mathbf{u}_i) \\ &= O_{\prec} \left(\frac{\|\mathbf{G}^{(i)}\|_F}{d} \right) \quad \text{(Khintchine's inequality [53])} \\ &= O_{\prec} \left(\frac{1}{\sqrt{d}} \right). \end{aligned} \tag{51}$$

Merging the results from (49) and (51) provides the relationship in Equation (48). ■

C.1 Proof of Lemma 2

For any $z \in \mathbb{C}$ with $\Im(z) > 0$, we define the same $\mathbf{G}(z)$, $\mathbf{G}^{(i)}$ and $s(z)$ as in Lemma 8. Denote $\tilde{\alpha} \stackrel{\text{def}}{=} \frac{n}{d-r}$, $m(z) \stackrel{\text{def}}{=} -s(z)$ and $m^{(i)}(z) \stackrel{\text{def}}{=} -s^{(i)}(z)$. Set $t = \frac{\alpha}{m(z)}$. We first show that

$$z = \psi(t) + O_{\prec} \left(\frac{1}{\sqrt{d}} \right). \quad (52)$$

Note that $(\mathbf{P} - z\mathbf{I})\mathbf{G}(z) = \mathbf{I}_{d-r}$. Taking the trace on both sides gives

$$\frac{1}{n} \sum_{i=1}^n y_i \mathbf{u}_i^\top \mathbf{G}(z) \mathbf{u}_i - z \text{Tr} \mathbf{G}(z) = d - r \quad (53)$$

For $\frac{1}{n} \mathbf{u}_i^\top \mathbf{G}(z) \mathbf{u}_i$, we use matrix inverse lemma to derive

$$\begin{aligned} \frac{1}{n} \mathbf{u}_i^\top \mathbf{G}(z) \mathbf{u}_i &= \frac{1}{n} \mathbf{u}_i^\top \left(\frac{1}{n} \sum_{j \neq i} y_j \mathbf{u}_j \mathbf{u}_j^\top - z\mathbf{I} + \frac{1}{n} y_i \mathbf{u}_i \mathbf{u}_i^\top \right)^{-1} \mathbf{u}_i \\ &= \frac{1}{n} \mathbf{u}_i^\top \left(\mathbf{G}^{(i)} - \frac{\frac{y_i}{n} \mathbf{G}^{(i)} \mathbf{u}_i \mathbf{u}_i^\top \mathbf{G}^{(i)}}{1 + \frac{y_i}{n} \mathbf{u}_i^\top \mathbf{G}^{(i)} \mathbf{u}_i} \right) \mathbf{u}_i \\ &= \frac{m^{(i)}(z)/\tilde{\alpha}}{y_i m^{(i)}(z)/\tilde{\alpha} - 1} \end{aligned}$$

Plug it back into (53) and denote $t^{(i)} = \frac{\tilde{\alpha}}{m^{(i)}(z)}$. We obtain:

$$\begin{aligned} &\frac{1}{d-r} \sum_{i=1}^n \frac{y_i/t^{(i)}}{y_i/t^{(i)} - 1} + z \cdot m(z) = 1 \\ \implies &\frac{\tilde{\alpha}}{m(z)} \cdot \frac{1}{n} \sum_{i=1}^n \frac{y_i}{y_i - t^{(i)}} + z = \frac{1}{\tilde{\alpha}} \cdot \frac{\tilde{\alpha}}{m(z)} \\ \implies &z = \frac{1}{\tilde{\alpha}} \cdot \frac{\tilde{\alpha}}{m(z)} + \frac{\tilde{\alpha}}{m(z)} \cdot \frac{1}{n} \sum_{i=1}^n \frac{y_i}{t^{(i)} - y_i}. \end{aligned}$$

By invoking Lemma 8 above, where $m(z) = m^{(i)}(z) + O_{\prec} \left(\frac{1}{\sqrt{d}} \right)$, it follows that $t = t^{(i)} + O_{\prec} \left(\frac{1}{\sqrt{d}} \right)$. Consequently,

$$\begin{aligned} z &= \frac{t}{\alpha} + \frac{1}{n} \sum_{i=1}^n \frac{ty_i}{t - y_i} + O_{\prec} \left(\frac{1}{\sqrt{d}} \right) \\ \implies z &= \frac{t}{\alpha} + \mathbb{E} \left[\frac{tY}{t - Y} \right] + O_{\prec} \left(\frac{1}{\sqrt{d}} \right) \\ \implies z &= \psi_{\alpha}(t) + O_{\prec} \left(\frac{1}{\sqrt{d}} \right). \end{aligned}$$

where the second step is due to the error bound on $\frac{1}{n} \sum_{i=1}^n \frac{ty_i}{t - y_i} - \mathbb{E} \left[\frac{tY}{t - Y} \right]$ (applying Theorem 7.7 in [44]). The identity $z = \psi(\alpha/m(z)) + O_{\prec} \left(\frac{1}{\sqrt{d}} \right)$ has been established for $z \in \mathbb{C}^+$. To extend this to a real spike eigenvalue $\tilde{\lambda}$, we rely on the established properties of the Stieltjes transform for large random matrices [52]. First, $m(z)$ converges almost surely to a deterministic limit $m^*(z)$, which is analytic outside the support of the limiting spectrum. As a spike, $\tilde{\lambda}$ lies outside this support, so $m^*(z)$ is continuous at $\tilde{\lambda}$. Second, the concentration bounds that give the $O_{\prec} \left(\frac{1}{\sqrt{d}} \right)$ error term remain valid for $z = \tilde{\lambda}$. This is because $\tilde{\lambda}$ is separated from the

spectrum of $\mathbf{P}$ by a spectral gap, ensuring that the resolvent $(\mathbf{P} - \tilde{\lambda}\mathbf{I})^{-1}$ is well-behaved. Consequently, $m(\tilde{\lambda})$ is well-defined and concentrates around its deterministic limit $m^*(\tilde{\lambda})$ such that $m(\tilde{\lambda}) = m^*(\tilde{\lambda}) + O_{\prec}\left(\frac{1}{\sqrt{d}}\right)$. By the continuity of ψ , it follows that $\psi(\alpha/m(\tilde{\lambda})) = \psi(\alpha/m^*(\tilde{\lambda})) + O_{\prec}\left(\frac{1}{\sqrt{d}}\right)$. Taking the limit $z \rightarrow \tilde{\lambda}$ in our established identity is therefore justified, completing the proof. $\blacksquare$

C.2 Proof of Lemma 3

To show the concentration of $\mathbf{M}(z) \stackrel{\text{def}}{=} \mathbf{A} + \mathbf{q}^\top (z\mathbf{I} - \mathbf{P})^{-1} \mathbf{q}$, we first show that $\mathbf{A}$ is concentrated on its limit $\mathbb{E}[\mathbf{Y}\mathbf{S}\mathbf{S}^\top]$. As $\mathbf{A}$ has finite dimensions, it suffices to derive the concentration of its individual entries, achievable through the application of Theorem 7.7 from [44]. Now it suffices to show the concentration of $\mathbf{q}^\top (z\mathbf{I} - \mathbf{P})^{-1} \mathbf{q}$, or equivalently, $\frac{1}{n} \mathbf{S} \mathbf{Y} \bar{\mathbf{U}}^\top (z\mathbf{I} - \bar{\mathbf{U}} \mathbf{Y} \bar{\mathbf{U}}^\top)^{-1} \bar{\mathbf{U}} \mathbf{Y} \mathbf{S}^\top$, where we define $\mathbf{q} \stackrel{\text{def}}{=} \frac{1}{\sqrt{n}} \bar{\mathbf{U}} \mathbf{Y} \mathbf{S}^\top$ and $\bar{\mathbf{U}} \stackrel{\text{def}}{=} \frac{1}{\sqrt{n}} \mathbf{U}$. Given the $r \times r$ dimension with r being finite, our analysis again reduces to each entry individually:

$$Q(\mathbf{a}, \mathbf{b}) \stackrel{\text{def}}{=} \mathbf{a}^\top \mathbf{Y} \bar{\mathbf{U}}^\top (z\mathbf{I} - \bar{\mathbf{U}} \mathbf{Y} \bar{\mathbf{U}}^\top)^{-1} \bar{\mathbf{U}} \mathbf{Y} \mathbf{b} \quad (54)$$

where $\mathbf{a}, \mathbf{b} \in \mathbb{R}^r$ are columns of $\frac{1}{\sqrt{n}} \mathbf{S}^\top$. Because $\mathbf{a}, \mathbf{b}$ have finite dimension r , it suffices to study (54) conditioning on $\mathbf{a}, \mathbf{b}$ and $\mathbf{Y}$ and only consider the randomness from $\bar{\mathbf{U}}$. An application of a union bound in the end completes the proof. Therefore, the rest of our proof progresses in two steps: first, we validate the concentration of (54) around its expectation; second, we characterize the expectation of $\mathbf{Y} \bar{\mathbf{U}}^\top (z\mathbf{I} - \bar{\mathbf{U}} \mathbf{Y} \bar{\mathbf{U}}^\top)^{-1} \bar{\mathbf{U}} \mathbf{Y}$ using some diagonal matrix $\mathbf{L}$. Combining the two steps provides the closeness of $\mathbf{M}(z)$ and $\mathbf{M}_{t^*}^*$.

Step 1. concentration of $Q(\mathbf{a}, \mathbf{b})$ at $\mathbb{E}Q(\mathbf{a}, \mathbf{b})$.

Observing from the form of (54), we are allowed to only focus on the submatrix of $\mathbf{Y}$ with $y_i \neq 0$. In other words, we can assume $\mathbf{Y}$ is invertible without loss of generality. This allows us to use Woodbury's matrix inverse lemma to derive

$$\mathbf{Y} \bar{\mathbf{U}}^\top (z\mathbf{I} - \bar{\mathbf{U}} \mathbf{Y} \bar{\mathbf{U}}^\top)^{-1} \bar{\mathbf{U}} \mathbf{Y} = \left(\mathbf{Y}^{-1} - \frac{1}{z} \bar{\mathbf{U}}^\top \bar{\mathbf{U}} \right)^{-1} - \mathbf{Y}. \quad (55)$$

As we conditioned on $\mathbf{a}, \mathbf{b}$ and $\mathbf{Y}$, it suffices to study

$$\left(\mathbf{Y}^{-1} - \frac{1}{z} \bar{\mathbf{U}}^\top \bar{\mathbf{U}} \right)^{-1} = \left(\mathbf{Y}^{-1} - \frac{1}{z} \sum_{i=1}^{d-r} \mathbf{v}_i \mathbf{v}_i^\top \right)^{-1}. \quad (56)$$

where we let $\mathbf{v}_i$ be the i -th column of $\bar{\mathbf{U}}^\top$.

Recall the classical Burkholder inequality (see Lemma 2.12 in [52]): Given a complex martingale difference sequence $\{r_k\}$ with respect to the increasing σ -field $\{\mathcal{F}_k\}$, for $p > 1$, we have

$$\mathbb{E} \left| \sum_k r_k \right|^p \leq C_p \mathbb{E} \left(\sum_k |r_k|^2 \right)^{p/2} \quad (57)$$

where C_p only depends on p . Now we construct the martingale difference sequence as follows. Denote

$$X \stackrel{\text{def}}{=} \mathbf{a}^\top \left(\mathbf{Y}^{-1} - \frac{1}{z} \sum_{i=1}^{d-r} \mathbf{v}_i \mathbf{v}_i^\top \right)^{-1} \mathbf{b}$$

and $\mathbb{E}_k$ as the operator to take the expectation with respect to $\mathbf{v}_1, \dots, \mathbf{v}_k$, with $\mathbb{E}_0 X = X$ and $\mathbb{E}_d X = \mathbb{E}[X]$. Our goal is to control the error

$$Q(\mathbf{a}, \mathbf{b}) - \mathbb{E}Q(\mathbf{a}, \mathbf{b}) = X - \mathbb{E}[X] = \sum_{k=1}^{d-r} (\mathbb{E}_{k-1} X - \mathbb{E}_k X) = \sum_{k=1}^{d-r} r_k. \quad (58)$$

where $r_k \stackrel{\text{def}}{=} \mathbb{E}_{k-1}X - \mathbb{E}_kX$. Then one can verify that $\{r_k\}$ is a martingale difference sequence. Our strategy is to bound the right side of (57) by some constant μ_p . Then Theorem 7.7 of [44] can be applied to bound the deviation $X - \mathbb{E}[X]$ using the notation of stochastic dominance. Therefore it remains to study the upper bound for the right side of (57).

To achieve that, we first show that $|r_k|^2$ is small. If we denote $\mathbf{G}_k = (\mathbf{Y}^{-1} - \frac{1}{z} \sum_{i \neq k} \mathbf{v}_i \mathbf{v}_i^\top)^{-1}$, then use Woodbury's matrix identity again, we get

$$\begin{aligned} X &= \mathbf{a}^\top \left(\mathbf{Y}^{-1} - \frac{1}{z} \sum_{i=1}^{d-r} \mathbf{v}_i \mathbf{v}_i^\top \right)^{-1} \mathbf{b} \\ &= \mathbf{a}^\top \left(\mathbf{Y}^{-1} - \frac{1}{z} \sum_{i \neq k} \mathbf{v}_i \mathbf{v}_i^\top - \frac{1}{z} \mathbf{v}_k \mathbf{v}_k^\top \right)^{-1} \mathbf{b} \\ &= \mathbf{a}^\top \left(\mathbf{G}_k + \frac{\mathbf{G}_k \mathbf{v}_k \mathbf{v}_k^\top \mathbf{G}_k / z}{1 - \mathbf{v}_k^\top \mathbf{G}_k \mathbf{v}_k / z} \right) \mathbf{b} \\ &= \mathbf{a}^\top \mathbf{G}_k \mathbf{b} + \frac{\mathbf{a}^\top \mathbf{G}_k \mathbf{v}_k \mathbf{v}_k^\top \mathbf{G}_k \mathbf{b}}{z - \mathbf{v}_k^\top \mathbf{G}_k \mathbf{v}_k}. \end{aligned}$$

The independence between $\mathbf{G}_k$ and $\mathbf{v}_k$ implies $\mathbb{E}_k[\mathbf{G}_k] = \mathbb{E}_{k-1}[\mathbf{G}_k]$. Hence

$$r_k = \mathbb{E}_{k-1}[X] - \mathbb{E}_k[X] = \mathbb{E}_{k-1} \left[\frac{\mathbf{a}^\top \mathbf{G}_k \mathbf{v}_k \mathbf{v}_k^\top \mathbf{G}_k \mathbf{b}}{z - \mathbf{v}_k^\top \mathbf{G}_k \mathbf{v}_k} \right] - \mathbb{E}_k \left[\frac{\mathbf{a}^\top \mathbf{G}_k \mathbf{v}_k \mathbf{v}_k^\top \mathbf{G}_k \mathbf{b}}{z - \mathbf{v}_k^\top \mathbf{G}_k \mathbf{v}_k} \right]$$

Furthermore,

$$|r_k| \leq \mathbb{E}_{k-1} \left[\frac{|\mathbf{a}^\top \mathbf{G}_k \mathbf{v}_k| |\mathbf{v}_k^\top \mathbf{G}_k \mathbf{b}|}{|z - \mathbf{v}_k^\top \mathbf{G}_k \mathbf{v}_k|} \right] + \mathbb{E}_k \left[\frac{|\mathbf{a}^\top \mathbf{G}_k \mathbf{v}_k| |\mathbf{v}_k^\top \mathbf{G}_k \mathbf{b}|}{|z - \mathbf{v}_k^\top \mathbf{G}_k \mathbf{v}_k|} \right]$$

We want to further simplify the right side. Similar to (55), we could derive the identity

$$\mathbf{G}_k = \mathbf{Y} + \mathbf{Y} \bar{\mathbf{U}}^{(k)\top} \left(z \mathbf{I} - \bar{\mathbf{U}}^{(k)} \mathbf{Y} \bar{\mathbf{U}}^{(k)\top} \right)^{-1} \bar{\mathbf{U}}^{(k)} \mathbf{Y}$$

where $\bar{\mathbf{U}}^{(k)} \in \mathbb{R}^{(d-r-1) \times n}$ is obtained by removing the k -th row of $\bar{\mathbf{U}}$. Write $z = E + i\eta$. One can verify that $\Im(-\mathbf{v}_k^\top \mathbf{G}_k \mathbf{v}_k)$ has the same sign as η . Hence, $|z - \mathbf{v}_k^\top \mathbf{G}_k \mathbf{v}_k| > |\Im(z - \mathbf{v}_k^\top \mathbf{G}_k \mathbf{v}_k)| \geq \eta$. As a result,

$$|r_k| \leq \frac{1}{|\eta|} \left(\mathbb{E}_{k-1}[|\mathbf{a}^\top \mathbf{G}_k \mathbf{v}_k| |\mathbf{v}_k^\top \mathbf{G}_k \mathbf{b}|] + \mathbb{E}_k[|\mathbf{a}^\top \mathbf{G}_k \mathbf{v}_k| |\mathbf{v}_k^\top \mathbf{G}_k \mathbf{b}|] \right)$$

Let $\alpha_k \stackrel{\text{def}}{=} |\mathbf{a}^\top \mathbf{G}_k \mathbf{v}_k|$ and $\beta_k \stackrel{\text{def}}{=} |\mathbf{v}_k^\top \mathbf{G}_k \mathbf{b}|$, then apply the AM-GM inequality $2xy \leq x^2 + y^2$ and Jensen's inequality on the convex function x^2 , we get

$$\begin{aligned} |r_k|^2 &\leq \frac{2}{\eta^2} \left((\mathbb{E}_{k-1}[\alpha_k \beta_k])^2 + (\mathbb{E}_k[\alpha_k \beta_k])^2 \right) \quad (\text{AM-GM inequality}) \\ &\leq \frac{2}{\eta^2} \left(\mathbb{E}_{k-1}[\alpha_k^2 \beta_k^2] + \mathbb{E}_k[\alpha_k^2 \beta_k^2] \right) \quad (\text{Jensen's inequality}) \\ &\leq \frac{1}{\eta^2} \left(\mathbb{E}_{k-1}[\alpha_k^4 + \beta_k^4] + \mathbb{E}_k[\alpha_k^4 + \beta_k^4] \right) \quad (\text{AM-GM inequality}). \end{aligned}$$

Now we have an upper bound for $|r_k|^2$. Recall that we aim to bound the expectation term on the right

side of (57). Consider an even p and write $q = \frac{p}{2}$. Then there exists K_p only depending on p such that:

$$\begin{aligned} \mathbb{E}\left(\sum_k |r_k|^2\right)^q &\leq \frac{1}{\eta^p} \mathbb{E}\left(\sum_k \mathbb{E}_{k-1} \alpha_k^4 + \sum_k \mathbb{E}_{k-1} \beta_k^4 + \sum_k \mathbb{E}_k \alpha_k^4 + \sum_k \mathbb{E}_k \beta_k^4\right)^q \\ &\leq \frac{K_p}{\eta^p} \left(\mathbb{E}\left(\sum_k \mathbb{E}_{k-1} \alpha_k^4\right)^q + \mathbb{E}\left(\sum_k \mathbb{E}_{k-1} \beta_k^4\right)^q + \mathbb{E}\left(\sum_k \mathbb{E}_k \alpha_k^4\right)^q + \mathbb{E}\left(\sum_k \mathbb{E}_k \beta_k^4\right)^q \right) \\ &\leq \frac{2K_p}{\eta^p} \left(\mathbb{E}\left(\sum_k \alpha_k^4\right)^q + \mathbb{E}\left(\sum_k \beta_k^4\right)^q \right), \end{aligned} \quad (59)$$

where the derivations are results of applying Jensen's inequality. We will focus on bounding $\mathbb{E}\left(\sum_k \alpha_k^4\right)^q$ and for $\mathbb{E}\left(\sum_k \beta_k^4\right)^q$ it is analogous. Apply Holder's inequality,

$$\mathbb{E}\left(\sum_k \alpha_k^4\right)^q = \mathbb{E} \sum_{k_1 \dots k_q} \alpha_{k_1}^4 \dots \alpha_{k_q}^4 \leq \sum_{k_1 \dots k_q} (\mathbb{E} \alpha_{k_1}^{4q})^{\frac{1}{q}} \dots (\mathbb{E} \alpha_{k_q}^{4q})^{\frac{1}{q}} = n^q \cdot \mathbb{E} \alpha_1^{4q}. \quad (60)$$

Recall $\alpha_1 \stackrel{\text{def}}{=} |\mathbf{a}^\top \mathbf{G}_1 \mathbf{v}_1|$ and $\mathbf{v}_1 \sim \mathcal{N}(0, \frac{1}{n} \mathbf{I})$ is defined to be the i -th column of $\overline{\mathbf{U}}^\top$. Condition on everything except for $\mathbf{v}_1$, we have $\mathbf{a}^\top \mathbf{G}_1 \mathbf{v}_1 \sim \mathcal{N}(0, \frac{\|\mathbf{a}^\top \mathbf{G}_1\|^2}{n})$. According to the L^p -norm bound of Gaussians (for example, see Section 2.5 of [54]), we have

$$\mathbb{E}_{v_1} \alpha_1^{4q} \leq A_q \frac{\|\mathbf{a}^\top \mathbf{G}_1\|^{4q}}{n^{2q}} \leq \frac{A_q}{n^{2q}} \|\mathbf{a}\|^{4q} \|\mathbf{G}_1\|^{4q} \quad (61)$$

for some value A_q only depending on q . Combining (60) with (61) and writing A_q as A_p , we get

$$\mathbb{E}\left(\sum_k \alpha_k^4\right)^q \leq \frac{A_p}{n^{p/2}} \|\mathbf{a}\|^{2p} \mathbb{E} \|\mathbf{G}_1\|^{2p}. \quad (62)$$

Similarly, there exists B_p such that

$$\mathbb{E}\left(\sum_k \beta_k^4\right)^q \leq \frac{B_p}{n^{p/2}} \|\mathbf{b}\|^{2p} \mathbb{E} \|\mathbf{G}_1\|^{2p}. \quad (63)$$

Now with all the building blocks, we are ready to go back and study $Q(\mathbf{a}, \mathbf{b})$. Combining the results from (57), (58), (59), (62) and (63) above, we have the following bound for the deviation of $Q(\mathbf{a}, \mathbf{b})$ from its mean:

$$\mathbb{E} |Q(\mathbf{a}, \mathbf{b}) - \mathbb{E} Q(\mathbf{a}, \mathbf{b})|^p \leq \frac{\kappa_p}{(\sqrt{n\eta})^p} (\|\mathbf{a}\|^{2p} + \|\mathbf{b}\|^{2p}) \mathbb{E} \|\mathbf{G}_1\|^{2p},$$

where κ_p only depends on p . Then it remains to control the term $\mathbb{E} \|\mathbf{G}_1\|^{2p}$.

Recall $\mathbf{G}_1 = \mathbf{Y} + \mathbf{Y} \overline{\mathbf{U}}^{(1)\top} \left(z \mathbf{I} - \overline{\mathbf{U}}^{(1)} \mathbf{Y} \overline{\mathbf{U}}^{(1)\top} \right)^{-1} \overline{\mathbf{U}}^{(1)} \mathbf{Y}$. Let $\tau = \max_{1 \leq i \leq n} |y_i|$. Then

$$\|\mathbf{G}_1\| \leq \tau + \tau^2 \|\overline{\mathbf{U}}\|^2 \|(z \mathbf{I} - \overline{\mathbf{U}} \mathbf{Y} \overline{\mathbf{U}}^\top)^{-1}\|$$

Denote $\{\lambda_i\}_{1 \leq i \leq d-r}$ the eigenvalues of $\overline{\mathbf{U}} \mathbf{Y} \overline{\mathbf{U}}^\top$ and recall $z = E + \iota \eta$. We have

$$\|\mathbf{G}_1\| \leq \tau + \tau^2 \|\overline{\mathbf{U}}\|^2 \max_{1 \leq i \leq d-r} \frac{1}{\sqrt{(E - \lambda_i)^2 + \eta^2}}.$$

For any $\delta > 0$, define $\mathbb{1}_{z,\delta} = \mathbb{1}_{\{\lambda_i \notin (E-\delta, E+\delta) \text{ for any } i\}}$. Then there exists constants C_p , k_p and $\tilde{k}_p$ that depend on p such that

$$\begin{aligned} \|\mathbf{G}_1\| &\leq \tau + \tau^2 \|\bar{\mathbf{U}}\|^2 \left(\frac{1}{\delta} \mathbb{1}_{z,\delta} + \frac{1}{|\eta|} \right) \\ \implies \|\mathbf{G}_1\|^{2p} &\leq k_p \left(\tau^{2p} + \frac{\tau^{4p} \|\bar{\mathbf{U}}\|^{4p} \mathbb{1}_{z,\delta}}{\delta^{2p}} + \frac{\tau^{4p} \|\bar{\mathbf{U}}\|^{4p}}{\eta^{2p}} \right) \\ \implies \mathbb{E} \|\mathbf{G}_1\|^{2p} &\leq k_p \left(\tau^{2p} + \frac{\tau^{4p} \tilde{k}_p}{\delta^{2p}} + \frac{\tau^{4p} C_p}{\eta^{2p}} \right) \end{aligned}$$

This final inequality shows that $\mathbb{E} \|\mathbf{G}_1\|^{2p}$ is bounded by a constant that depends on p , τ , δ , and η , but crucially, it does not grow with the dimension d . This establishes the necessary moment bounds for the martingale difference sequence, which is sufficient for our main stochastic domination argument to hold.

Step 2. show $\|\mathbb{E}[\mathbf{Y}\bar{\mathbf{U}}(z\mathbf{I} - \bar{\mathbf{U}}\mathbf{Y}\bar{\mathbf{U}}^\top)^{-1}\bar{\mathbf{U}}\mathbf{Y}] - \mathbf{L}\| = O_{\prec}\left(\frac{1}{\sqrt{d}}\right)$. Define $m^*(z)$ as the unique solution satisfying

$$z = \frac{1}{\alpha} \cdot \frac{\alpha}{m^*(z)} + \frac{\alpha}{m^*(z)} \cdot \mathbb{E} \left[\frac{Y}{\frac{\alpha}{m^*(z)} - Y} \right] \quad (64)$$

From the proof of Lemma 2, we know that

$$z = \frac{1}{\tilde{\alpha}} \cdot \frac{\tilde{\alpha}}{m(z)} + \frac{\tilde{\alpha}}{m(z)} \cdot \frac{1}{n} \sum_{i=1}^n \frac{y_i}{\frac{\tilde{\alpha}}{m(z)} - y_i}. \quad (65)$$

and $m^*(z) - m(z) = O_{\prec}\left(\frac{1}{\sqrt{d}}\right)$. Thus, $m^*(z)$ is the limiting version of $m(z)$, which implies that $\frac{\alpha}{m^*(z)}$ is the limit of $\frac{\alpha}{m(z)}$. By Lemmas 1 and 2, we know $\frac{\alpha}{m(z)}$ converges to the solutions to the fix-point equations (12) and (13). Hence any

$$t^* \stackrel{\text{def}}{=} \frac{\alpha}{m^*(z)} \in \{t_{1,u}^*, \dots, t_{r,u}^*\} \cup \{t_{1,l}^*, \dots, t_{r,l}^*\}. \quad (66)$$

Define diagonal matrices

$$\mathbf{L} \stackrel{\text{def}}{=} \mathbb{E} \left[\frac{Y^2}{t^* - Y} \right] \cdot \mathbf{I} \quad \text{and} \quad \tilde{\mathbf{L}} \stackrel{\text{def}}{=} \text{diag} \left\{ \frac{y_i^2}{\frac{\tilde{\alpha}}{m(z)} - y_i} \right\}.$$

Our second step of the proof of Lemma 3 aims to bound the difference $\|\mathbb{E}[\mathbf{Y}\bar{\mathbf{U}}(z\mathbf{I} - \bar{\mathbf{U}}\mathbf{Y}\bar{\mathbf{U}}^\top)^{-1}\bar{\mathbf{U}}\mathbf{Y}] - \mathbf{L}\|$. To achieve this, it suffices to study $\|\mathbb{E}[\mathbf{Y}\bar{\mathbf{U}}(z\mathbf{I} - \bar{\mathbf{U}}\mathbf{Y}\bar{\mathbf{U}}^\top)^{-1}\bar{\mathbf{U}}\mathbf{Y}] - \tilde{\mathbf{L}}\|$. Furthermore, it will be more convenient to study $\mathbf{Y}\bar{\mathbf{U}}(z\mathbf{I} - \bar{\mathbf{U}}\mathbf{Y}\bar{\mathbf{U}}^\top)^{-1}\bar{\mathbf{U}}\mathbf{Y} + \mathbf{Y}$, which equals to $(\mathbf{Y}^{-1} - \frac{1}{z}\bar{\mathbf{U}}^\top\bar{\mathbf{U}})^{-1}$. Since

$$\tilde{\mathbf{L}} + \mathbf{Y} = \text{diag} \left\{ \frac{\frac{\tilde{\alpha}}{m(z)} y_i}{\frac{\tilde{\alpha}}{m(z)} - y_i} \right\} = \left(\mathbf{Y}^{-1} - \frac{m(z)}{\tilde{\alpha}} \mathbf{I} \right)^{-1},$$

our goal is to bound the difference

$$\left\| \mathbb{E} \left(\mathbf{Y}^{-1} - \frac{1}{z} \bar{\mathbf{U}}^\top \bar{\mathbf{U}} \right)^{-1} - \left(\mathbf{Y}^{-1} - \frac{m(z)}{\tilde{\alpha}} \mathbf{I} \right)^{-1} \right\|. \quad (67)$$

Applying the identity $\mathbf{A}^{-1} - \mathbf{B}^{-1} = \mathbf{A}^{-1}(\mathbf{B} - \mathbf{A})\mathbf{B}^{-1}$, we get

$$\begin{aligned} &\left(\mathbf{Y}^{-1} - \frac{1}{z} \bar{\mathbf{U}}^\top \bar{\mathbf{U}} \right)^{-1} - \left(\mathbf{Y}^{-1} - \frac{m(z)}{\tilde{\alpha}} \mathbf{I} \right)^{-1} \\ &= \left(\mathbf{Y}^{-1} - \frac{1}{z} \bar{\mathbf{U}}^\top \bar{\mathbf{U}} \right)^{-1} \left(\frac{1}{z} \bar{\mathbf{U}}^\top \bar{\mathbf{U}} - \frac{m(z)}{\tilde{\alpha}} \mathbf{I} \right) \left(\mathbf{Y}^{-1} - \frac{m(z)}{\tilde{\alpha}} \mathbf{I} \right)^{-1} \\ &= \mathbf{G} \left(\frac{1}{z} \bar{\mathbf{U}}^\top \bar{\mathbf{U}} - \frac{m(z)}{\tilde{\alpha}} \mathbf{I} \right) \left(\mathbf{Y}^{-1} - \frac{m(z)}{\tilde{\alpha}} \mathbf{I} \right)^{-1} \end{aligned} \quad (68)$$

if we denote $\mathbf{G} = \left(\mathbf{Y}^{-1} - \frac{1}{z} \overline{\mathbf{U}}^\top \overline{\mathbf{U}}\right)^{-1}$. Recall we let $\mathbf{v}_i$ be the i -th column of $\overline{\mathbf{U}}^\top$ and $\mathbf{G}_i = (\mathbf{Y}^{-1} - \frac{1}{z} \sum_{j \neq i} \mathbf{v}_j \mathbf{v}_j^\top)^{-1}$. Previously, we derived

$$\mathbf{G} = \mathbf{G}_i + \frac{\frac{1}{z} \mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top \mathbf{G}_i}{1 - \frac{1}{z} \mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i} = \mathbf{G}_i + \frac{\mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top \mathbf{G}_i}{z - \mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i}.$$

Multiply $\mathbf{v}_i$ on the right, we have

$$\mathbf{G} \mathbf{v}_i = \frac{z}{z - \mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i} \mathbf{G}_i \mathbf{v}_i.$$

As a result, the product of the first two terms in (68) boils down to:

$$\begin{aligned} \mathbf{G} \left(\frac{1}{z} \overline{\mathbf{U}}^\top \overline{\mathbf{U}} - \frac{m(z)}{\tilde{\alpha}} \mathbf{I} \right) &= \sum_{i=1}^{d-r} \frac{\mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top}{z - \mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i} - \mathbf{G} \cdot \frac{m(z)}{\tilde{\alpha}} \\ &= \sum_{i=1}^{d-r} \frac{\mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top}{z - \mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i} - \frac{1}{d-r} \sum_{i=1}^{d-r} \left(\mathbf{G}_i + \frac{\mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top \mathbf{G}_i}{z - \mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i} \right) \frac{m(z)}{\tilde{\alpha}} \\ &= \sum_{i=1}^{d-r} \frac{\mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top}{z - \mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i} - \frac{1}{n} \sum_{i=1}^{d-r} \left(\mathbf{G}_i + \frac{\mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top \mathbf{G}_i}{z - \mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i} \right) m(z). \end{aligned} \quad (69)$$

Note that $\mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i = \frac{1}{n} \text{Tr} \mathbf{G}_i + O_{\prec} \left(\frac{1}{\sqrt{d}} \right) = \frac{1}{n} \sum_{j=1}^n \frac{\frac{\tilde{\alpha}}{m(z)} y_j}{\frac{\tilde{\alpha}}{m(z)} - y_j} + O_{\prec} \left(\frac{1}{\sqrt{d}} \right)$, where the first equality can be derived similar as in the analysis for the quadratic form in (51) and the second equality follows from the fixed-point characterization of the Stieltjes transform $m(z)$ established in the proof of Lemma 2, combined with the fact that $(1/n) \text{Tr} \mathbf{G}_i = m(z) + O_{\prec} \left(\frac{1}{\sqrt{d}} \right)$. Now denote $\delta_i \stackrel{\text{def}}{=} \mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i - \frac{1}{n} \sum_{j=1}^n \frac{\frac{\tilde{\alpha}}{m(z)} y_j}{\frac{\tilde{\alpha}}{m(z)} - y_j}$. We can write

$$\frac{1}{z - \mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i} = \frac{1}{z - \frac{1}{n} \sum_{j=1}^n \frac{\frac{\tilde{\alpha}}{m(z)} y_j}{\frac{\tilde{\alpha}}{m(z)} - y_j} - \delta_i} = \frac{1}{\frac{1}{m(z)} - \delta_i} = \frac{m(z)}{1 - m(z) \delta_i} = m(z) + \frac{\delta_i m^2(z)}{1 - m(z) \delta_i}. \quad (70)$$

where the second equality is due to (65). Substitute this identity into the expression (69), we obtain

$$\begin{aligned} \mathbf{G} \left(\frac{1}{z} \overline{\mathbf{U}}^\top \overline{\mathbf{U}} - \frac{m(z)}{\tilde{\alpha}} \mathbf{I} \right) &= \sum_{i=1}^{d-r} \left(m(z) + \frac{\delta_i m^2(z)}{1 - m(z) \delta_i} \right) \mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top - \frac{1}{n} \sum_{i=1}^{d-r} \mathbf{G}_i m(z) \\ &\quad - \frac{m(z)}{n} \sum_{i=1}^{d-r} \left(m(z) + \frac{\delta_i m^2(z)}{1 - m(z) \delta_i} \right) \mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top \mathbf{G}_i \end{aligned}$$

Note that $\mathbf{G}_i$ is independent of $\mathbf{v}_i$ and $\mathbb{E} \mathbf{v}_i \mathbf{v}_i^\top = \frac{1}{n} \mathbf{I}$. As a result,

$$\mathbb{E} \left[\mathbf{G} \left(\frac{1}{z} \overline{\mathbf{U}}^\top \overline{\mathbf{U}} - \frac{m(z)}{\tilde{\alpha}} \mathbf{I} \right) \right] = \mathbb{E} \left[\sum_{i=1}^{d-r} \frac{\delta_i m^2(z)}{1 - m(z) \delta_i} \mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top - \frac{m^2(z)}{n^2} \sum_{i=1}^{d-r} \mathbf{G}_i^2 - \frac{m(z)}{n} \sum_{i=1}^{d-r} \frac{\delta_i m^2(z)}{1 - m(z) \delta_i} \mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top \mathbf{G}_i \right]$$

Applying $\|\mathbb{E} \mathbf{A}\| \leq \mathbb{E} \|\mathbf{A}\|$ and the triangle inequality, we proceed to control the three terms:

$$\begin{aligned} \Delta_1 &\stackrel{\text{def}}{=} \mathbb{E} \left\| \sum_{i=1}^{d-r} \frac{\delta_i m^2(z)}{1 - m(z) \delta_i} \mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top \right\| \\ \Delta_2 &\stackrel{\text{def}}{=} \mathbb{E} \left\| \frac{m^2(z)}{n^2} \sum_{i=1}^{d-r} \mathbf{G}_i^2 \right\| \\ \Delta_3 &\stackrel{\text{def}}{=} \mathbb{E} \left\| \frac{m(z)}{n} \sum_{i=1}^{d-r} \frac{\delta_i m^2(z)}{1 - m(z) \delta_i} \mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top \mathbf{G}_i \right\| \end{aligned}$$

For Δ_1 , recall that we have $\mathbf{G}\mathbf{v}_i = \frac{z}{z - \mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i} \mathbf{G}_i \mathbf{v}_i$ and $\frac{\delta_i m^2(z)}{1 - m(z)\delta_i} = \frac{\delta_i m(z)}{z - \mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i}$ (induced from equating the first and the fourth terms in (70) above). Therefore,

$$\begin{aligned} \sum_{i=1}^{d-r} \frac{\delta_i m^2(z)}{1 - m(z)\delta_i} \mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top &= \sum_{i=1}^{d-r} \frac{\delta_i m(z)}{z} \mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top = \frac{m(z)\mathbf{G}}{z} \sum_{i=1}^{d-r} \delta_i \mathbf{v}_i \mathbf{v}_i^\top \\ \Rightarrow \left\| \sum_{i=1}^{d-r} \frac{\delta_i m^2(z)}{1 - m(z)\delta_i} \mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top \right\| &\leq \frac{\|\mathbf{G}\| |m(z)|}{|z|} \cdot \max_{1 \leq i \leq d-r} |\delta_i| \cdot \left\| \sum_{i=1}^{d-r} \mathbf{v}_i \mathbf{v}_i^\top \right\|. \end{aligned}$$

Recall the bounded support for Y (by assumption in (8)) and for the limiting spectrum of the Marchenko Pastur law. Then it is easy to show that $\|\mathbf{G}\| = O_{\prec}(1)$, $m(z) = O_{\prec}(1)$ and $\mathbf{v}_i^\top \mathbf{G}_i \mathbf{v}_i = O_{\prec}\left(\frac{1}{\sqrt{d}}\right)$. Consequently, $\max_{1 \leq i \leq d-r} |\delta_i| = O_{\prec}\left(\frac{1}{\sqrt{d}}\right)$ and both $\frac{\|\mathbf{G}\| |m(z)|}{|z|}$ and $\left\| \sum_{i=1}^{d-r} \mathbf{v}_i \mathbf{v}_i^\top \right\|$ are $O_{\prec}(1)$. For Δ_2 , we have

$$\mathbb{E} \left\| \frac{m^2(z)}{n^2} \sum_{i=1}^{d-r} \mathbf{G}_i^2 \right\| \leq \frac{|m^2(z)|}{n^2} \sum_{i=1}^{d-r} \mathbb{E} \|\mathbf{G}_i\| = O_{\prec}\left(\frac{1}{n}\right) = O_{\prec}\left(\frac{1}{d}\right).$$

For Δ_3 ,

$$\mathbb{E} \left\| \frac{m(z)}{n} \sum_{i=1}^{d-r} \frac{\delta_i m^2(z)}{1 - m(z)\delta_i} \mathbf{G}_i \mathbf{v}_i \mathbf{v}_i^\top \mathbf{G}_i \right\| \leq \frac{|m(z)|}{n} \cdot \sum_{i=1}^{d-r} \frac{|\delta_i| |m^2(z)|}{|1 - m(z)\delta_i|} \cdot \|\mathbf{G}_i \mathbf{v}_i\|^2$$

and one can verify that $\frac{|m(z)|}{n} = O_{\prec}\left(\frac{1}{d}\right)$, $\sum_{i=1}^{d-r} \frac{|\delta_i| |m^2(z)|}{|1 - m(z)\delta_i|} = O_{\prec}\left(\frac{1}{\sqrt{d}}\right)$ and $\|\mathbf{G}_i \mathbf{v}_i\|^2 = O_{\prec}(1)$. Hence we have shown that all the three terms Δ_1 , Δ_2 , and Δ_3 have order $O_{\prec}\left(\frac{1}{\sqrt{d}}\right)$. Therefore,

$$\left\| \mathbb{E} \left[\mathbf{G} \left(\frac{1}{z} \bar{\mathbf{U}}^\top \bar{\mathbf{U}} - \frac{m(z)}{\alpha} \right) \right] \right\| = O_{\prec}\left(\frac{1}{\sqrt{d}}\right).$$

Note the norm of the term $\left(\mathbf{Y}^{-1} - \frac{m(z)}{\alpha} \mathbf{I} \right)^{-1}$ in (68) has order $O_{\prec}(1)$. This provides a bound for (67) and eventually leads to

$$\|\mathbb{E}[\mathbf{Y} \bar{\mathbf{U}} (z\mathbf{I} - \bar{\mathbf{U}} \mathbf{Y} \bar{\mathbf{U}}^\top)^{-1} \bar{\mathbf{U}} \mathbf{Y}] - \mathbf{L}\| = O_{\prec}\left(\frac{1}{\sqrt{d}}\right).$$

Finally, combining the two main steps of our proof, we have derived the order of the deviation

$$\mathbf{Y} \bar{\mathbf{U}} (z\mathbf{I} - \bar{\mathbf{U}} \mathbf{Y} \bar{\mathbf{U}}^\top)^{-1} \bar{\mathbf{U}} \mathbf{Y} - \mathbb{E} \left[\frac{Y^2}{t^* - Y} \right] \cdot \mathbf{I},$$

where $t^* = \frac{\alpha}{m^*(z)}$, and moreover, the matrix

$$\mathbf{M}(z) = \mathbf{A} + \mathbf{q}^\top (z\mathbf{I} - \mathbf{P})^{-1} \mathbf{q} = \mathbf{A} + \frac{1}{n} \mathbf{S} \mathbf{Y} \bar{\mathbf{U}}^\top (z\mathbf{I} - \bar{\mathbf{U}} \mathbf{Y} \bar{\mathbf{U}}^\top)^{-1} \bar{\mathbf{U}} \mathbf{Y} \mathbf{S}^\top$$

converges to

$$\mathbf{M}_{t^*}^* = \mathbb{E}[\mathbf{Y} \mathbf{S} \mathbf{S}^\top] + \mathbb{E} \left[\frac{Y}{t^* - Y} \mathbf{S} \mathbf{S}^\top \right].$$

This leads to (29). Furthermore, because both matrices have finite dimensions $r \times r$. The difference between their corresponding eigenvalues would also be $O_{\prec}\left(\frac{1}{\sqrt{d}}\right)$. Therefore (30) is obtained. This completes the proof of Lemma 3. $\blacksquare$

D Lemmas for Eigenvector Characterizations

Lemma 9 *With the same notations in Lemma 6, $(z\mathbf{I} - \mathbf{M}(z))^{-1}$ is well-defined and analytic inside γ except at λ .*

proof: When $z \in \mathbb{R}$, it is clear that any z that makes $z\mathbf{I} - \mathbf{M}(z)$ non-invertible is an eigenvalue of $\mathbf{M}(z)$. Consequently, our focus shifts to the cases where $\Im(z) \neq 0$. Denote $\rho \stackrel{\text{def}}{=} \Re(z), \eta \stackrel{\text{def}}{=} \Im(z)$. Considering the eigendecomposition $\mathbf{P} = \sum_{k=1}^{d-k} \lambda_k \mathbf{u}_k \mathbf{u}_k^\top$, we have

$$\mathbf{q}^\top (z\mathbf{I} - \mathbf{P})^{-1} \mathbf{q} = \mathbf{q}^\top \left(\sum_{k=1}^{d-r} \frac{1}{\rho + \iota\eta - \lambda_k} \mathbf{u}_k \mathbf{u}_k^\top \right) \mathbf{q} = \sum_{k=1}^{d-r} \frac{\rho - \lambda_k - \iota\eta}{(\rho - \lambda_k)^2 + \eta^2} \mathbf{q}^\top \mathbf{u}_k \mathbf{u}_k^\top \mathbf{q}.$$

Define $\mathbf{E}(z) \stackrel{\text{def}}{=} \Re(z\mathbf{I} - \mathbf{M}(z)), \mathbf{\Gamma}(z) \stackrel{\text{def}}{=} \Im(z\mathbf{I} - \mathbf{M}(z))$. It follows that

$$\mathbf{\Gamma}(z) = \eta \mathbf{I} + \sum_{k=1}^{d-r} \frac{\eta}{(\rho - \lambda_k)^2 + \eta^2} \mathbf{q}^\top \mathbf{u}_k \mathbf{u}_k^\top \mathbf{q},$$

indicating that $\mathbf{\Gamma}(z) \succ \mathbf{0}$ if $\eta > \mathbf{0}$ and $\mathbf{\Gamma}(z) \prec \mathbf{0}$ otherwise. To show the non-singularity of $(z\mathbf{I} - \mathbf{M}(z))$, consider any vector $\mathbf{v} = \mathbf{a} + \iota \mathbf{b}$ such that $(z\mathbf{I} - \mathbf{M}(z))\mathbf{v} = \mathbf{0}$. We proceed to show that $\mathbf{v}$ must be the zero vector.

Note that $(z\mathbf{I} - \mathbf{M}(z))\mathbf{v} = \mathbf{0}$ implies $\mathbf{E}(z)\mathbf{a} - \mathbf{\Gamma}(z)\mathbf{b} = \mathbf{0}$ and $\mathbf{E}(z)\mathbf{b} + \mathbf{\Gamma}(z)\mathbf{a} = \mathbf{0}$, which further implies the system of equations:

$$\begin{aligned} \mathbf{b}^\top \mathbf{E} \mathbf{a} &= \mathbf{b}^\top \mathbf{\Gamma}(z) \mathbf{b}, \\ \mathbf{a}^\top \mathbf{E} \mathbf{b} &= -\mathbf{a}^\top \mathbf{\Gamma}(z) \mathbf{a}. \end{aligned}$$

Subtracting to get

$$\mathbf{b}^\top \mathbf{\Gamma}(z) \mathbf{b} + \mathbf{a}^\top \mathbf{\Gamma}(z) \mathbf{a} = 0. \quad (71)$$

For $\eta > 0$, $\mathbf{\Gamma}(z)$ is a positive definite matrix. Thus $\mathbf{a} = \mathbf{b} = \mathbf{0}$ is necessary for (71) to hold. For the case when $\eta < 0$ it is analogous. Hence, we conclude that $\mathbf{v} = \mathbf{0}$, which completes the proof. $\blacksquare$

Proof of Lemma 7 Without loss of generality, let λ^* be the leading eigenvalue of $\mathbf{M}_{t^*}^*$. As $z \rightarrow \lambda^*$ and $\zeta'_\alpha(\lambda^*) > 0$, it follows that $\zeta'_\alpha(z) > 0$ for sufficiently large d . By focusing on the segment of ϕ where its derivative is positive, we can define its inverse function and let $\tilde{t}$ be the value $\zeta_\alpha^{-1}(z)$. For any $t > \tau_u$, let $\mathbf{M}_t^* = \mathbf{A}^* + \mathbf{J}$ where $\mathbf{A}^* \stackrel{\text{def}}{=} \mathbb{E}[\mathbf{Y} \mathbf{S} \mathbf{S}^\top]$ and $\mathbf{J} \stackrel{\text{def}}{=} \mathbb{E}[\frac{\mathbf{Y}^2}{t-Y} \mathbf{S} \mathbf{S}^\top]$ and consider the eigendecomposition

$$\mathbf{M}_{t^*}^* = \sum_{\lambda_j \neq \lambda^*} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top + \lambda^* \sum_{\lambda_j = \lambda^*} \mathbf{v}_j \mathbf{v}_j^\top = \sum_{\lambda_j \neq \lambda^*} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top + \lambda^* \sum_{1 \leq j \leq k} \mathbf{v}_j \mathbf{v}_j^\top$$

Let $\delta = z - \lambda^*$ and $w = \tilde{t} - t^*$. Then

$$\begin{aligned} z\mathbf{I} - \mathbf{M}^*(z) &= z\mathbf{I} - \mathbf{M}^*(\lambda^*) + \mathbf{M}^*(\lambda^*) - \mathbf{M}^*(z) \\ &= \sum_{\lambda_j \neq \lambda^*} (\delta + \lambda^* - \lambda_j) \mathbf{u}_j \mathbf{u}_j^\top + \delta \sum_{1 \leq j \leq k} \mathbf{v}_j \mathbf{v}_j^\top + \mathbf{M}_{t^*}^* - \mathbf{M}_t^* \\ &= \sum_{\lambda_j \neq \lambda^*} (\delta + \lambda^* - \lambda_j) \mathbf{u}_j \mathbf{u}_j^\top + \delta \sum_{1 \leq j \leq k} \mathbf{v}_j \mathbf{v}_j^\top + \mathbb{E} \left[Y^2 \left(\frac{1}{t^* - Y} - \frac{1}{\tilde{t} - Y} \right) \mathbf{S} \mathbf{S}^\top \right] \end{aligned}$$

Using Taylor expansion, we get

$$\frac{1}{\tilde{t} - Y} = \frac{1}{w + t^* - Y} = \frac{1}{t^* - Y} - \frac{w}{(t^* - Y)^2} + w^2 \Delta,$$

where $\Delta = \frac{1}{(t^* - Y)^3} - \frac{w}{(t^* - Y)^4} + \frac{w^2}{(t^* - Y)^5} - \dots$. As a result,

$$\begin{aligned}
z\mathbf{I} - \mathbf{M}^*(z) &= \sum_{\lambda_j \neq \lambda^*} (\delta + \lambda^* - \lambda_j) \mathbf{u}_j \mathbf{u}_j^\top + \delta \sum_{1 \leq j \leq k} \mathbf{v}_j \mathbf{v}_j^\top + \mathbb{E} \left[Y^2 w \left(\frac{1}{(t^* - Y)^2} - w\Delta \right) \mathbf{S} \mathbf{S}^\top \right] \\
&= \sum_{\lambda_j \neq \lambda^*} (\delta + \lambda^* - \lambda_j) \mathbf{u}_j \mathbf{u}_j^\top + \delta \sum_{1 \leq j \leq k} \mathbf{v}_j \mathbf{v}_j^\top + w \left(\mathbb{E} \left[Y^2 \frac{1}{(t^* - Y)^2} \mathbf{S} \mathbf{S}^\top \right] + w\bar{\Delta} \right),
\end{aligned}$$

where $\bar{\Delta} \stackrel{\text{def}}{=} \mathbb{E} \left[-Y^2 \Delta \mathbf{S} \mathbf{S}^\top \right]$. Now denote $\tilde{\mathbf{E}} \stackrel{\text{def}}{=} \mathbb{E} \left[Y^2 \left(\frac{1}{(t^* - Y)^2} - w\Delta \right) \mathbf{S} \mathbf{S}^\top \right]$ and $\mathbf{U} = [\mathbf{u}_1 \dots \mathbf{u}_{r-k} \mathbf{v}_1 \dots \mathbf{v}_k] \in \mathbb{R}^{r \times r}$, $\bar{\mathbf{U}} = [\mathbf{u}_1 \dots \mathbf{u}_{r-k}] \in \mathbb{R}^{r \times (r-k)}$ (eigenvectors not associated with λ^*) and $\bar{\mathbf{V}} = [\mathbf{v}_1 \dots \mathbf{v}_k] \in \mathbb{R}^{r \times k}$ (eigenvectors associated with λ^*). Then

$$z\mathbf{I} - \mathbf{M}^*(z) = \mathbf{U} \begin{bmatrix} \mathbf{C} & \mathbf{F}^\top \\ \mathbf{F} & \mathbf{G} \end{bmatrix} \mathbf{U}^\top$$

where

$$\begin{cases} \mathbf{C} & \stackrel{\text{def}}{=} \text{diag}\{\delta + \lambda^* - \lambda_j\} + w\bar{\mathbf{U}}^\top \tilde{\mathbf{E}} \bar{\mathbf{U}} \in \mathbb{R}^{(r-k) \times (r-k)}, \\ \mathbf{F} & \stackrel{\text{def}}{=} w\bar{\mathbf{V}} \tilde{\mathbf{E}} \bar{\mathbf{U}} \in \mathbb{R}^{k \times (r-k)}, \\ \mathbf{G} & \stackrel{\text{def}}{=} \delta \mathbf{I} + w\bar{\mathbf{V}}^\top \tilde{\mathbf{E}} \bar{\mathbf{V}} \in \mathbb{R}^{k \times k}. \end{cases}$$

Apply the formula for inverse of 2×2 block matrices [45]:

$$(z\mathbf{I} - \mathbf{M}^*(z))^{-1} = \mathbf{U} \begin{bmatrix} \mathbf{H} & \mathbf{N}^\top \\ \mathbf{N} & \mathbf{K} \end{bmatrix} \mathbf{U}^\top$$

where

$$\begin{cases} \mathbf{H} & \stackrel{\text{def}}{=} (\mathbf{C} - \mathbf{F}^\top \mathbf{G}^{-1} \mathbf{F})^{-1} \in \mathbb{R}^{(r-k) \times (r-k)}, \\ \mathbf{N} & \stackrel{\text{def}}{=} -\mathbf{G}^{-1} \mathbf{F} \mathbf{H} \in \mathbb{R}^{k \times (r-k)}, \\ \mathbf{K} & \stackrel{\text{def}}{=} \mathbf{G}^{-1} + \mathbf{G}^{-1} \mathbf{F} \mathbf{H} \mathbf{F}^\top \mathbf{G}^{-1} \in \mathbb{R}^{k \times k}. \end{cases}$$

Note that by mean value theorem, there exists t between $\tilde{t}$ and t^* such that $\delta = \zeta_\alpha'(t)w$. Then we could employ matrix inverse lemma on $\mathbf{G}^{-1} = (\delta \mathbf{I} + w\bar{\mathbf{V}} \tilde{\mathbf{E}} \bar{\mathbf{V}})^{-1}$ to write $\mathbf{F}^\top \mathbf{G}^{-1} \mathbf{F}$ as $\delta \tilde{\mathbf{G}}$, where $\tilde{\mathbf{G}}$ is some matrix that converges to a constant matrix when $\delta \rightarrow 0$. Consequently, $\mathbf{F}^\top \mathbf{G}^{-1} \mathbf{F} \rightarrow \mathbf{0}$. Apply similar analysis, we will get $\delta \mathbf{H} \rightarrow \mathbf{0}$, $\delta \mathbf{N} \rightarrow \mathbf{0}$ and $\delta(\mathbf{K} - \mathbf{G}^{-1}) \rightarrow \mathbf{0}$. Therefore, we can derive the residue as

$$(z - \lambda^*) (z\mathbf{I} - \mathbf{M}^*(z))^{-1} = \delta (z\mathbf{I} - \mathbf{M}^*(z))^{-1} \rightarrow \mathbf{V}^* \left(\mathbf{I} + \mathbf{V}^{*\top} \frac{\mathbf{E}(t^*)}{\zeta_\alpha'(t^*)} \mathbf{V}^* \right)^{-1} \mathbf{V}^{*\top}$$

where $\mathbf{E}(t^*) = \mathbb{E} \left[\frac{Y^2}{(t^* - Y)^2} \mathbf{S} \mathbf{S}^\top \right]$ is the limit of $\tilde{\mathbf{E}}$ as $\delta \rightarrow 0$. ■

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